

The Sorcerer
13,000 BC

The deep roots of human biology II

Dr Laura van Holstein

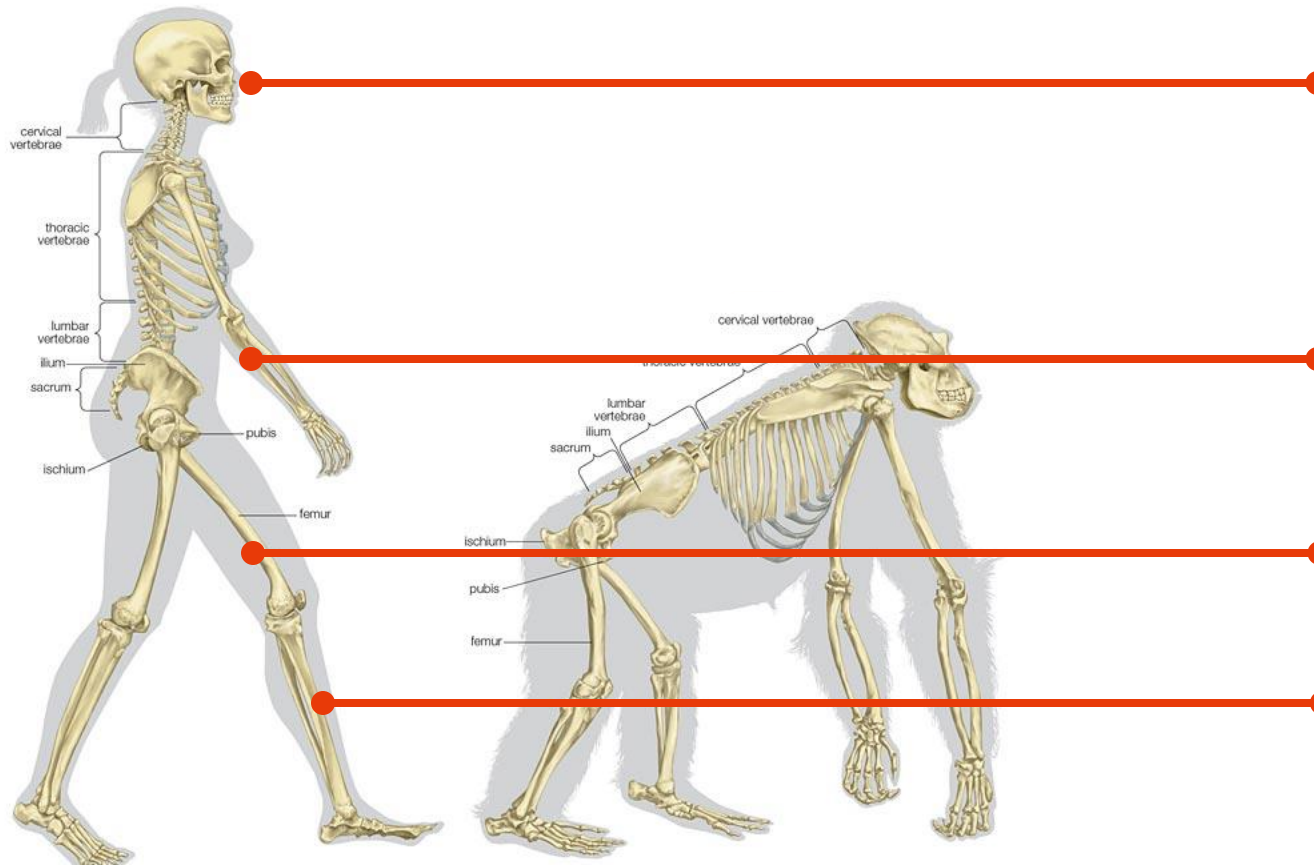
lavanho@emory.edu

Anthropology 215

Office hours: Wednesdays 9-10.30AM



Picking up where we left off...



Cranial characters

- **Chin**
- **Globular cranium**
- High EQ
- **Gracile face positioned underneath cranium**
- No Cp3 honing complex

Postcrania: general descriptors

- **Gracile**
- Exceptionally plastic (?)

Postcrania: body proportions

- Long legs, short arms

Postcrania: bipedal adaptations

- Long legs, short arms
- Bowl-shaped pelvis
- Elongated femur
- Flexible lower back
- Highly derived feet
- Inferiorly placed foramen magnum

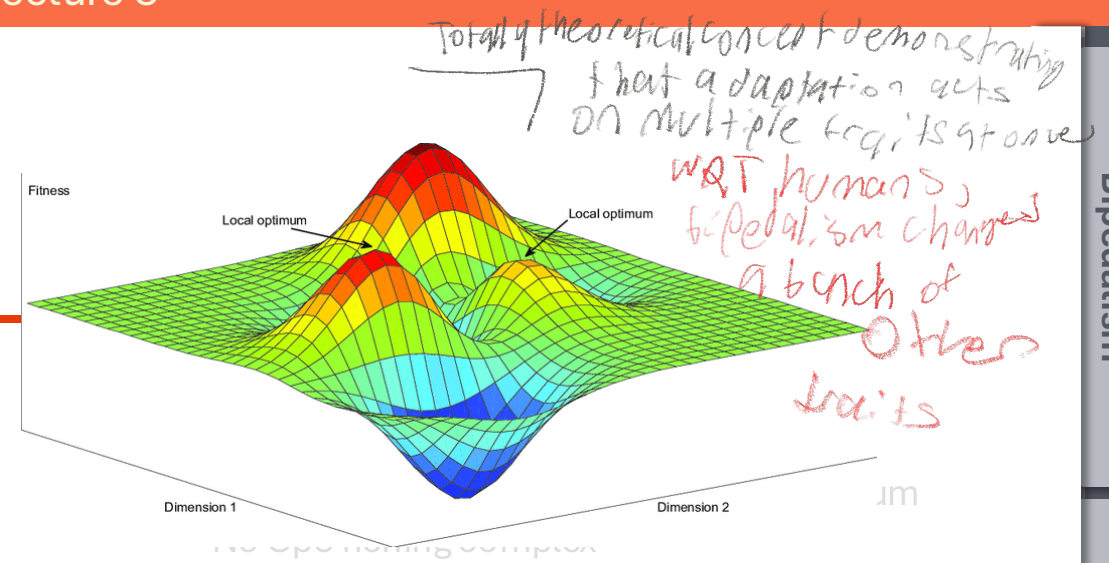
Picking up where we left off...



Bipedalism is a key trait

- Totally reorganises most of the postcranial skeleton: to understand the **why** of modern human **morphology** (@ med students!) you must understand bipedalism
 - Necessitates changes in every part of adaptive strategy
- Underlies dispersal capacity of the species & necessitates longer childhood
 - Plasticity?
- Underlies modern human physiology (basal metabolic rate)

able to walk longer distances



Postcrania: general descriptors

- Gracile
- Exceptionally plastic (?)

Postcrania: body proportions

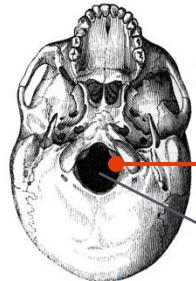
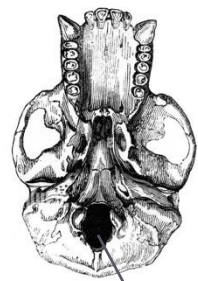
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Postcrania: bipedal adaptations

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Bipedalism reorganizes the entire postcranial skeleton

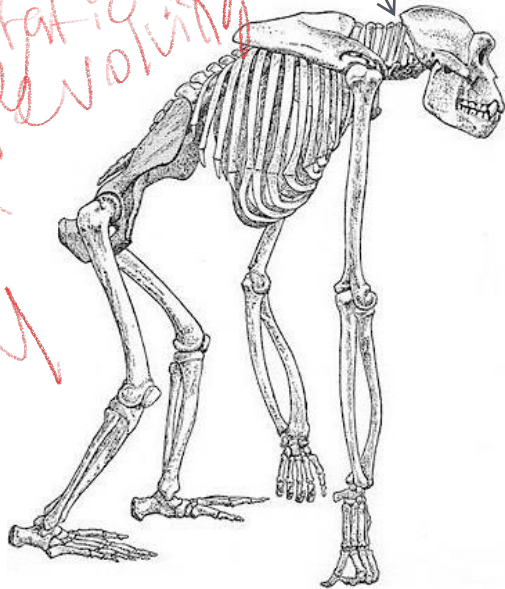


Chimps are good model for MRCA —
 Select pressures lead to reorganization
 around bipedalism

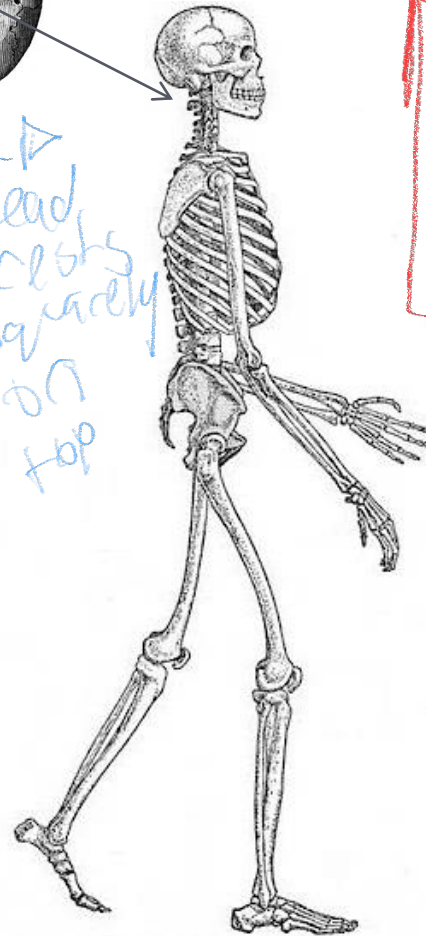
• **Foramen magnum** inferoanteriorly placed

Hole in the skull spinal
 cord passes through

Chimps
 weren't
 static
 been evolving
 for
 my



Head
 sits
 squarely
 on
 top



But this is problematic,
 MRCA potentially had
 upright body posture
 and likely moved similarly to
 orangutan

Bipedalism reorganizes the entire postcranial skeleton

Foramen magnum inferoanteriorly placed

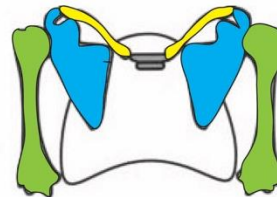
Shoulder Scapulae oriented so that humerus on side



Top View

Front View

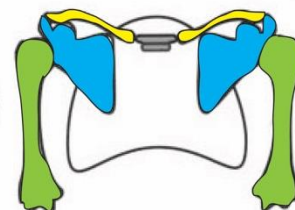
Right Side View



Top View

Front View

Right Side View



Chimpanzee shoulder

Chest is wide front-to-back
Scapula is placed on the back
Clavicles are long and angled
Humerus on the side of the rib cage



Homo sapiens shoulder

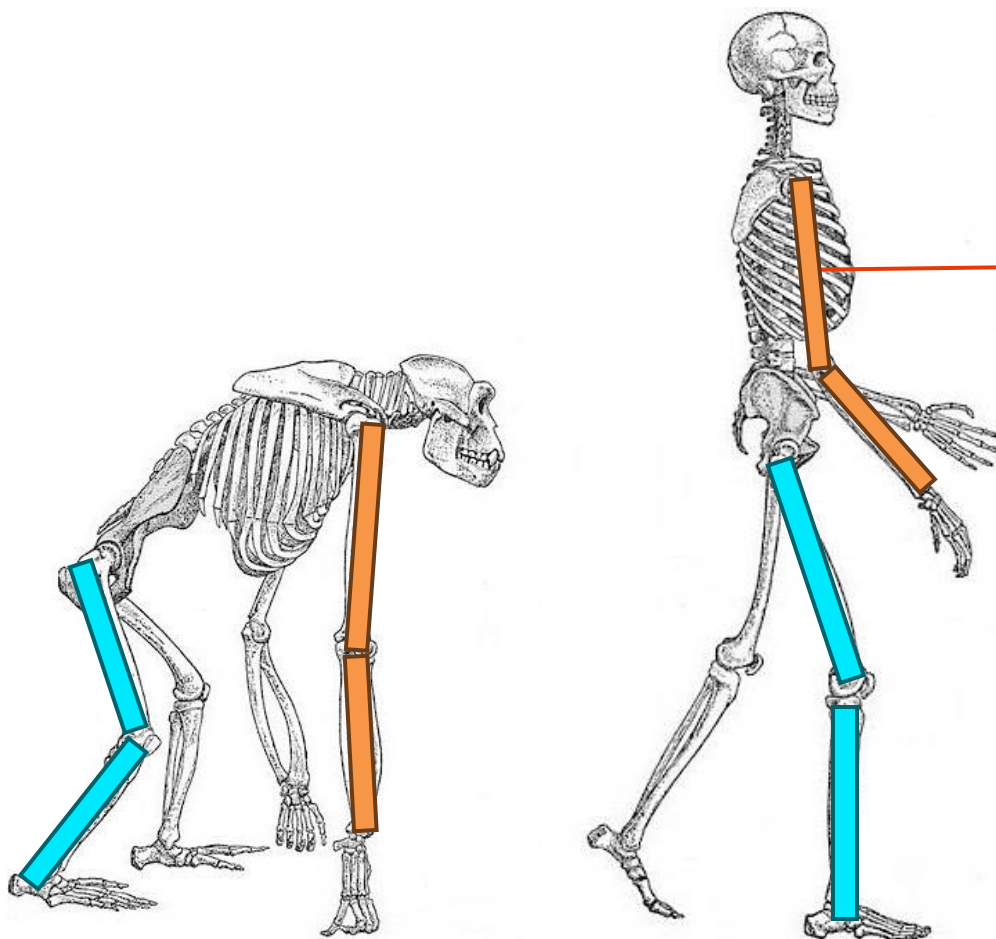
Chest is wide front-to-back
Scapula is placed on the back
Clavicles are *short and horizontal*
Humerus is placed on the side of our rib cage so we can pull our arms back to throw and run

Bipedalism reorganizes the entire postcranial skeleton

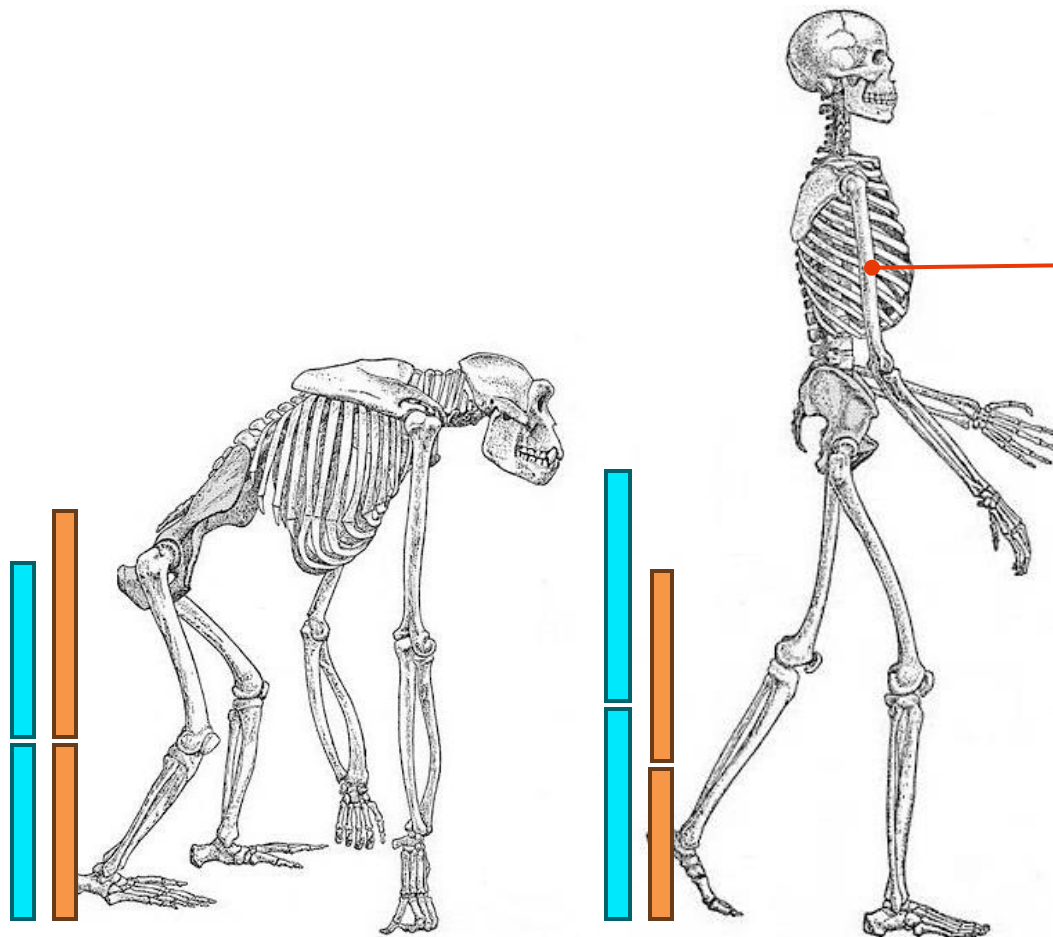
Foramen magnum inferoanteriorly placed

Shoulder Scapulae oriented so that humerus on side

• **Arms** Short, relative to body and legs



Bipedalism reorganizes the entire postcranial skeleton



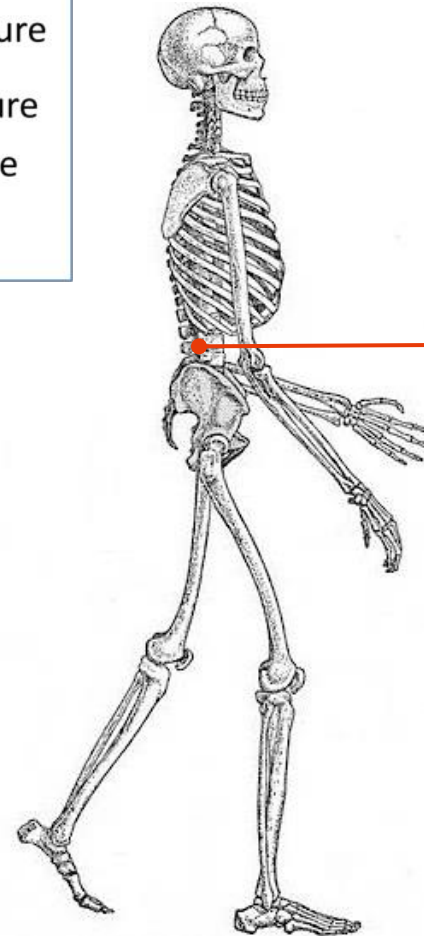
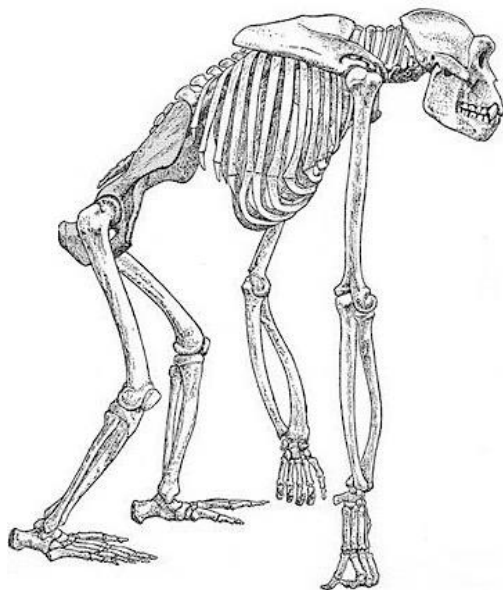
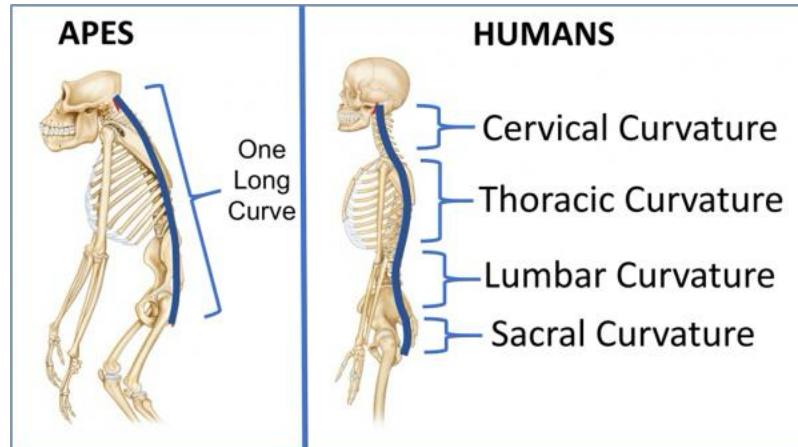
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• **Arms** Short, relative to body and legs

Humans have much longer legs, + w.r.t length of arms are even longer

Bipedalism reorganizes the entire postcranial skeleton



Foramen magnum inferoanteriorly placed

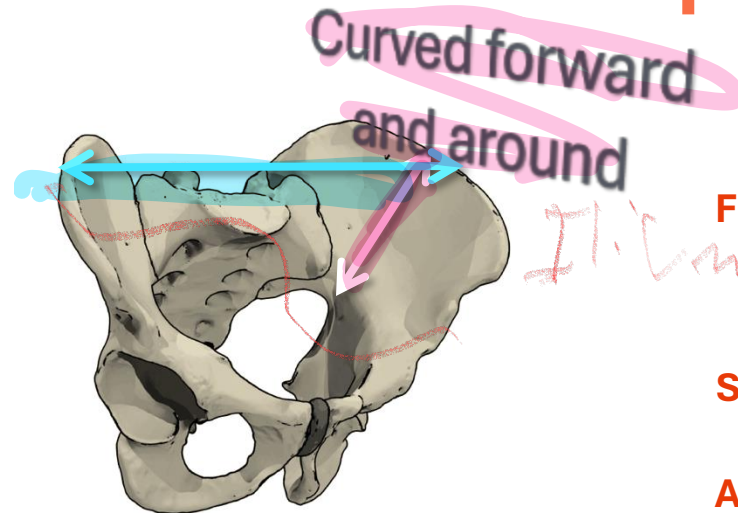
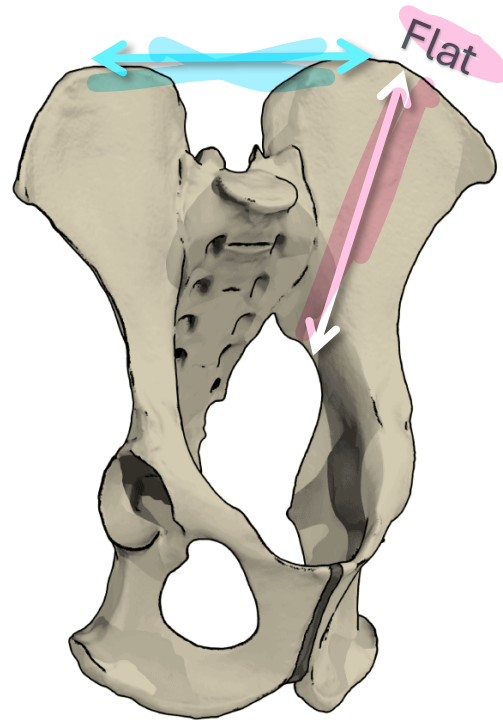
Shoulder Scapulae oriented so that humerus on side

Arms Short, relative to body and legs

• **Lumbar spine** Flexible; lordosis (S-shaped spine)

Flexible lumbar region to keep upright when walking or else we move side to side which is energetically expensive

Bipedalism reorganizes the entire postcranial skeleton



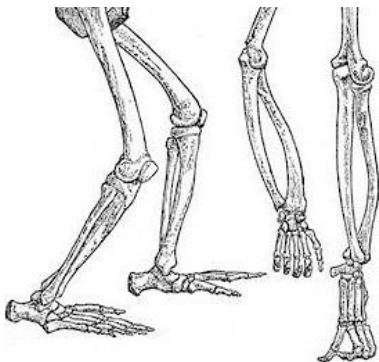
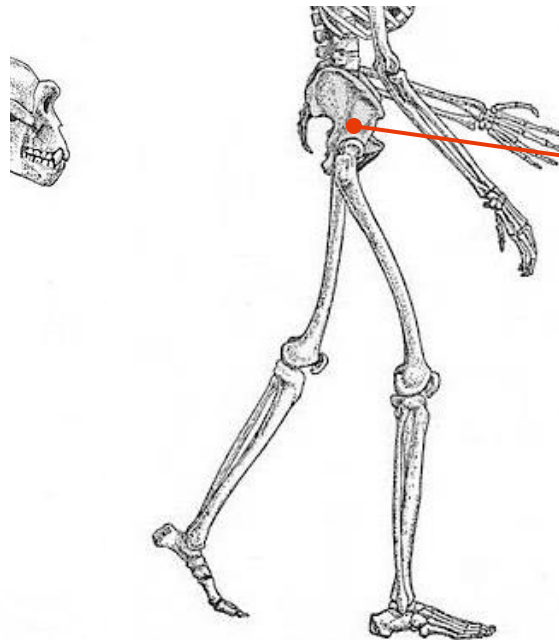
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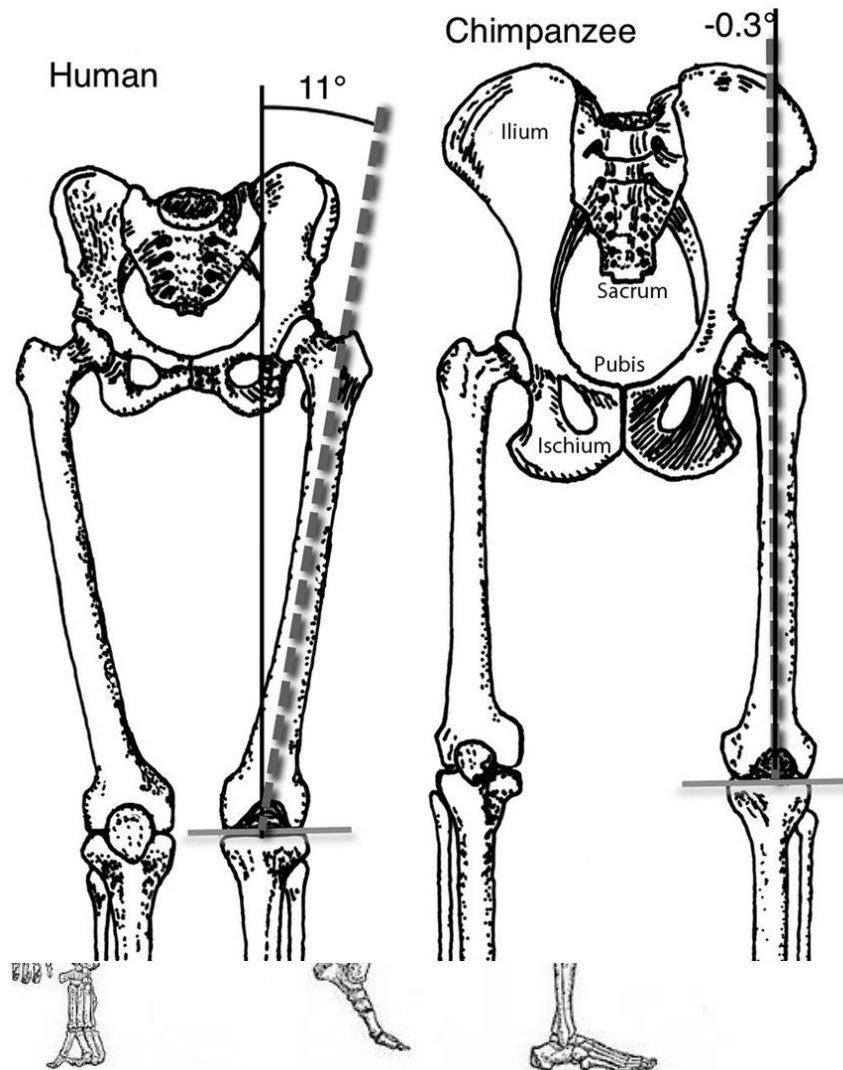
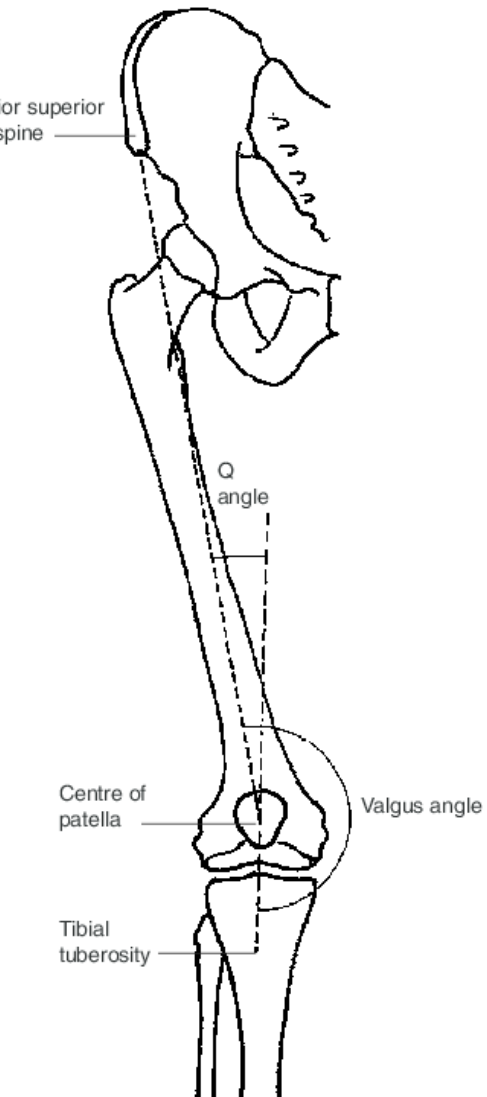
Arms Short, relative to body and legs

Lumbar spine Flexible; lordosis (S-shaped spine)

Pelvis 'Bowl-shaped'. Ilium supero-inferiorly shortened and sagittally oriented. Mediolaterally broad.



Bipedalism reorganizes the entire postcranial skeleton



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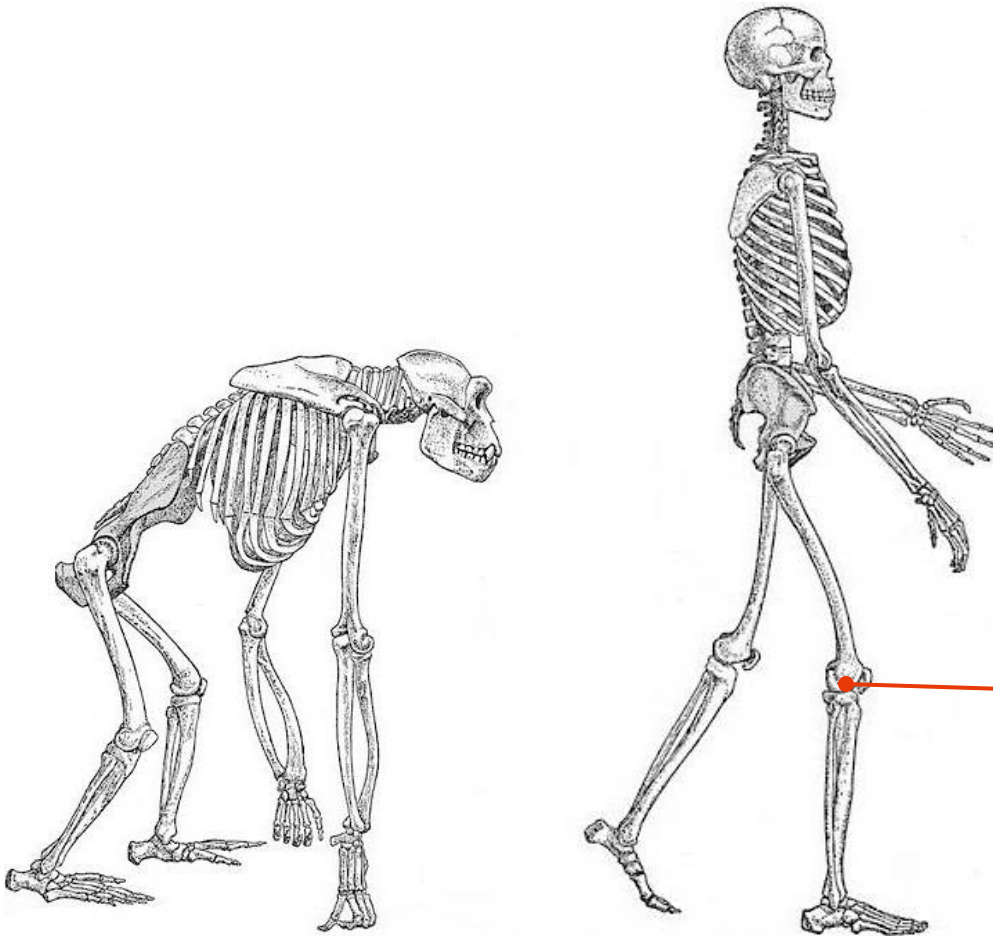
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→ **Legs** Long relative to body and arms. Q and valgus angle

Bipedalism reorganizes the entire postcranial skeleton



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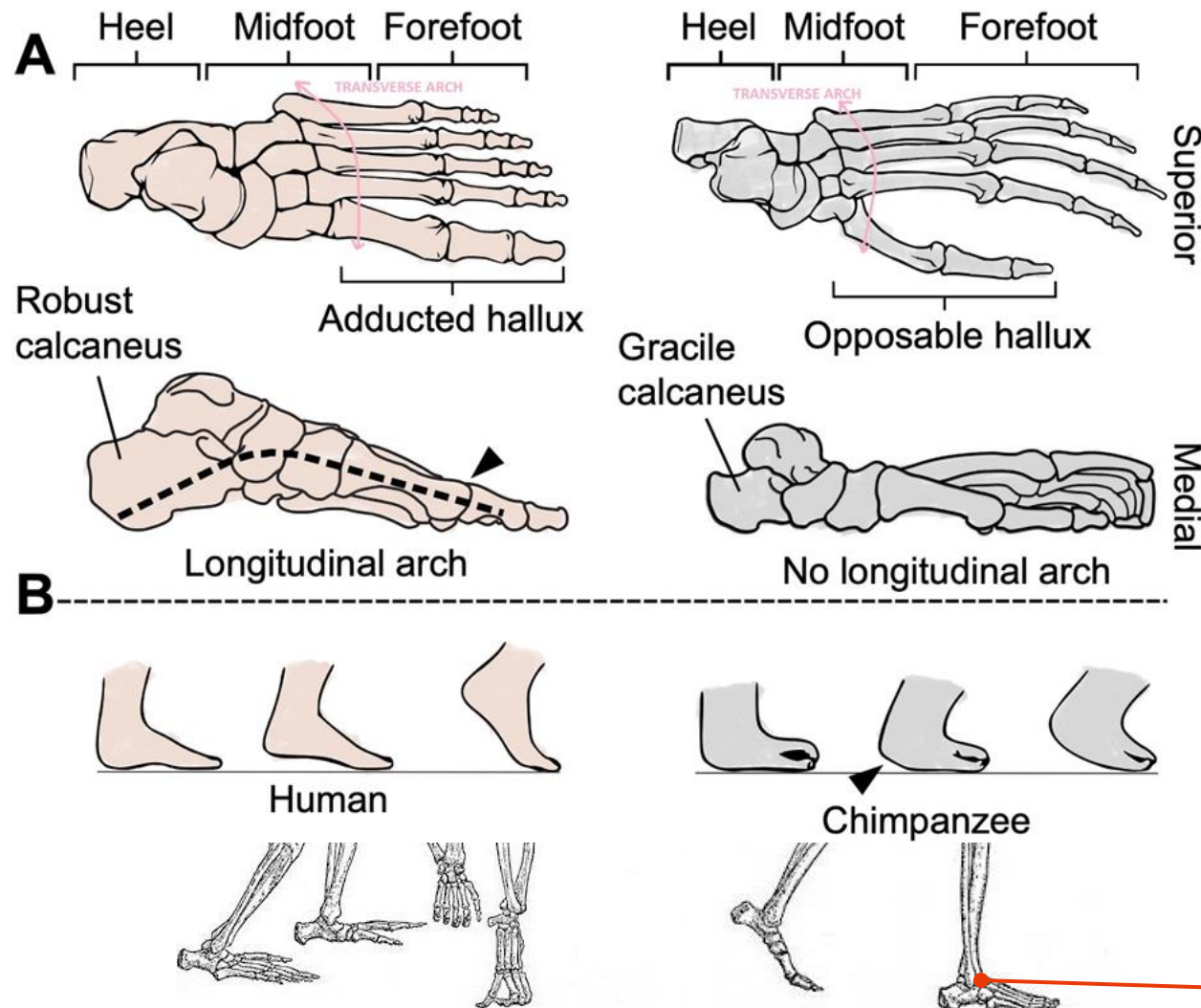
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• **Leg joints** Larger than in chimpanzees

Bipedalism reorganizes the entire postcranial skeleton



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Feet Stiff midfoot; large calcaneus; adducted hallux

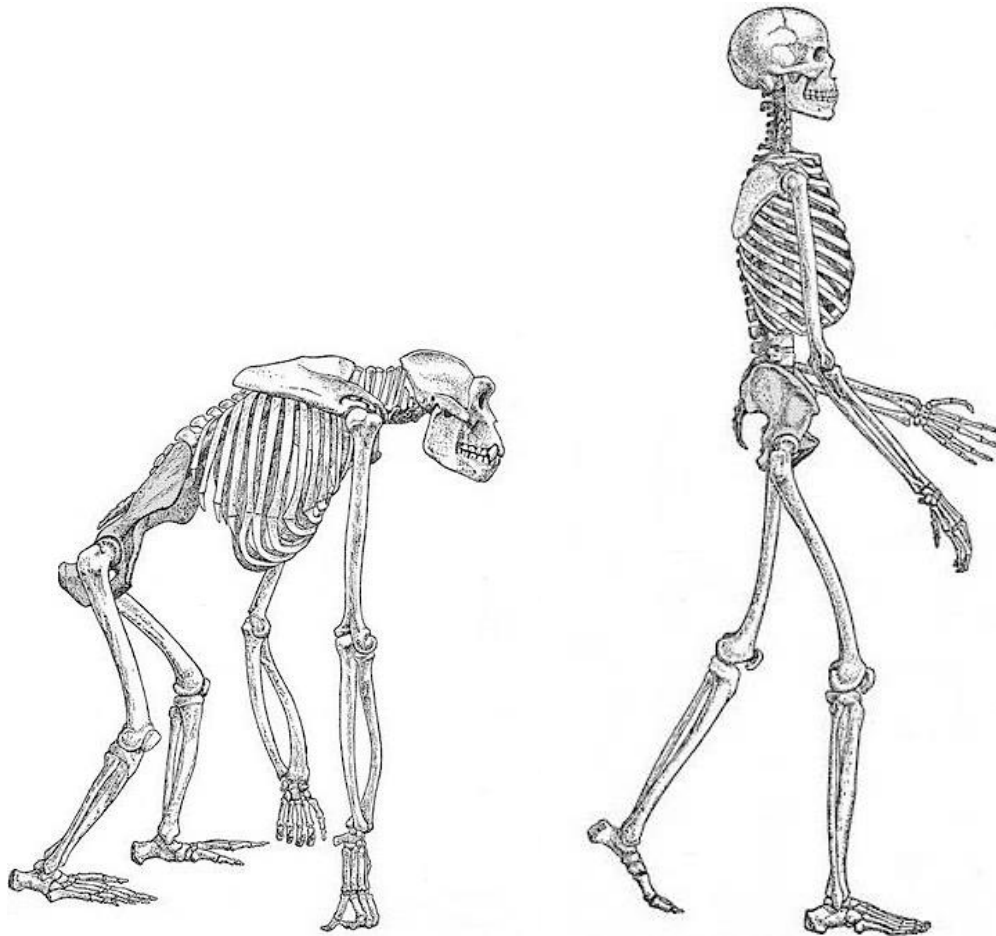
Bipedalism reorganizes the entire postcranial skeleton

*This suite of traits
is oriented towards bipedalism*

Bipedalism

Why a biped?

Consequences



EFFICIENT BIPEDAL LOCOMOTION

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Arms Short, relative to body and legs

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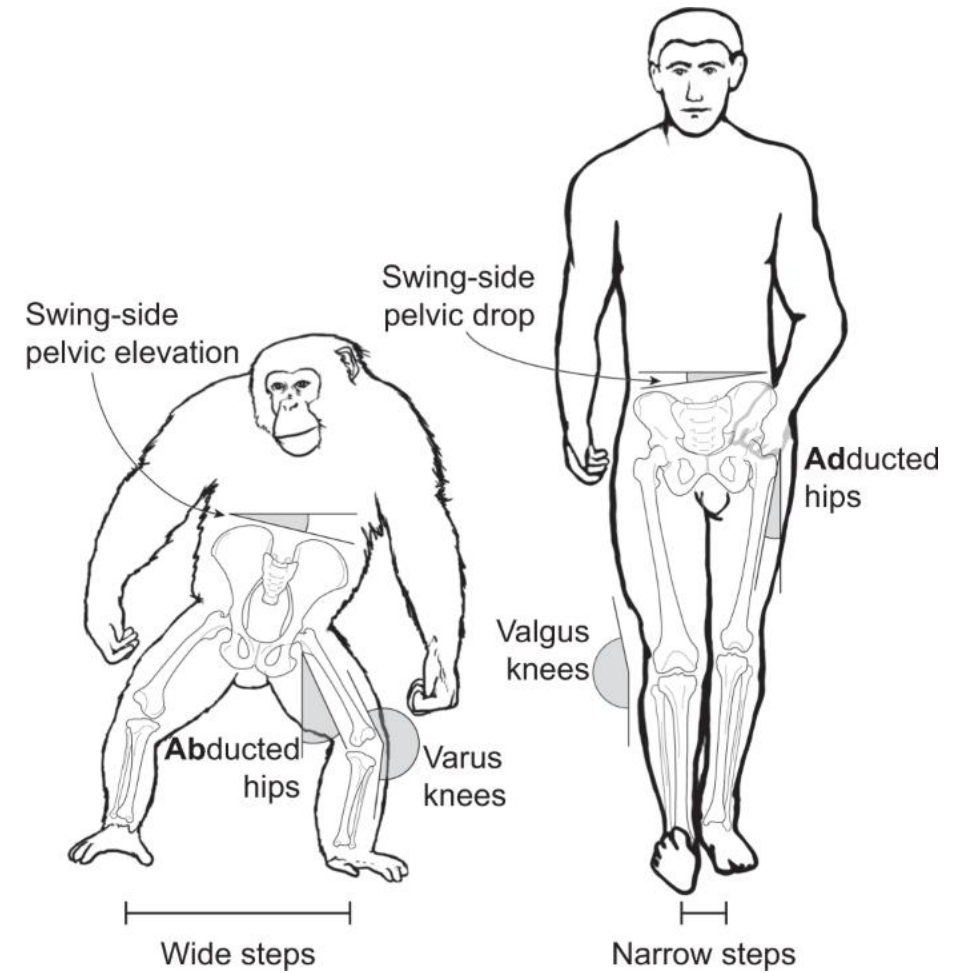
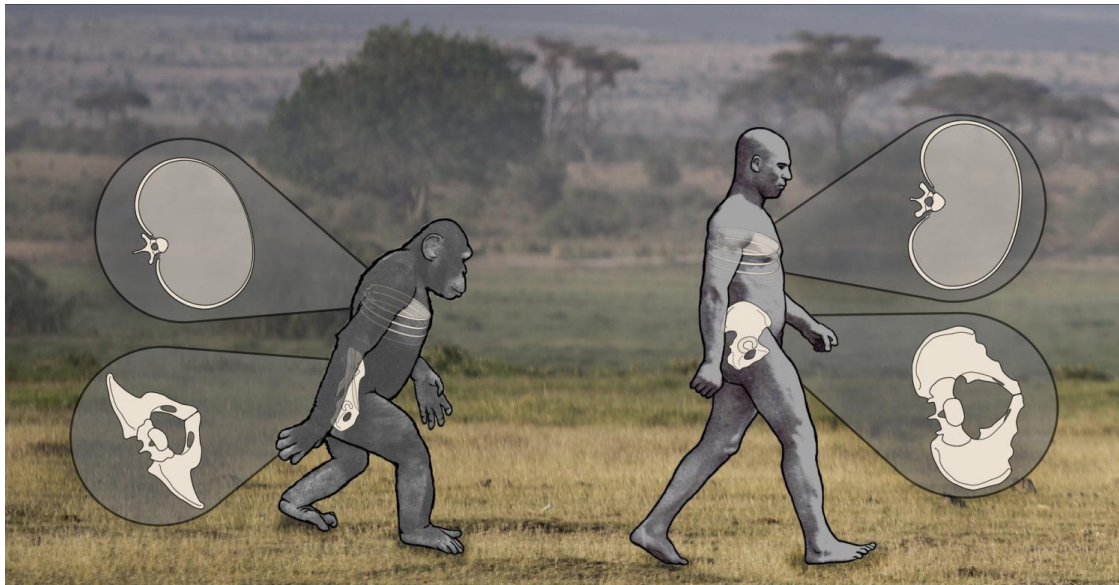
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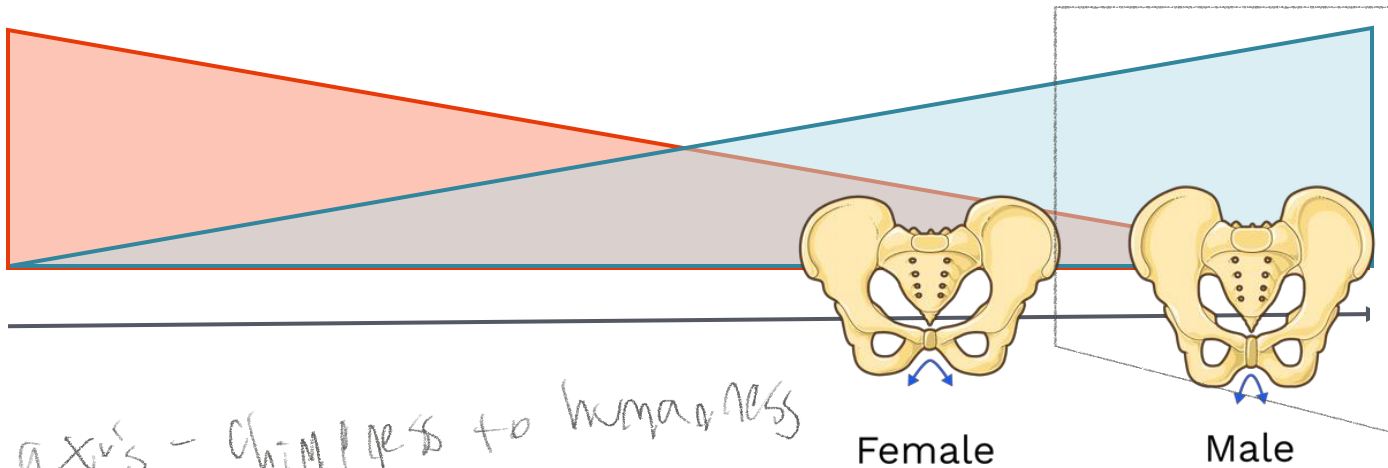


Bipedalism reorganizes the entire postcranial skeleton ...which entails huge tradeoffs

axis cost

human tradeoff

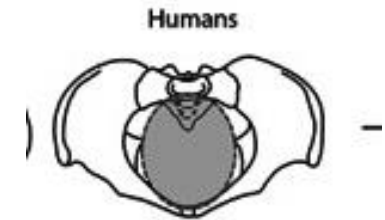
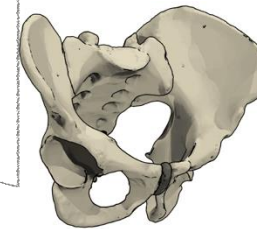
COST



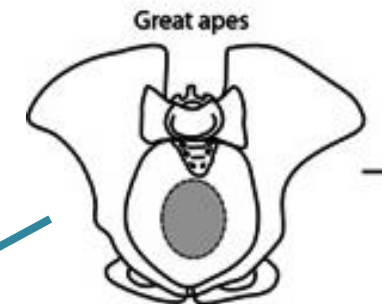
Female

Male

Bipedal locomotor efficiency
Constraints on neonate size
Difficult childbirth

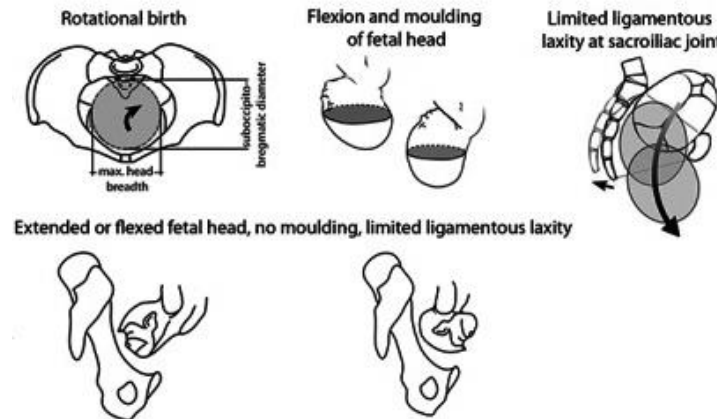
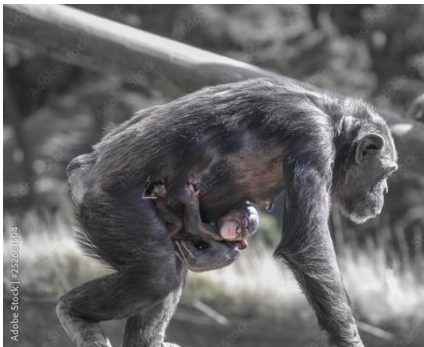


Humans



Great apes

Secondary altriciality

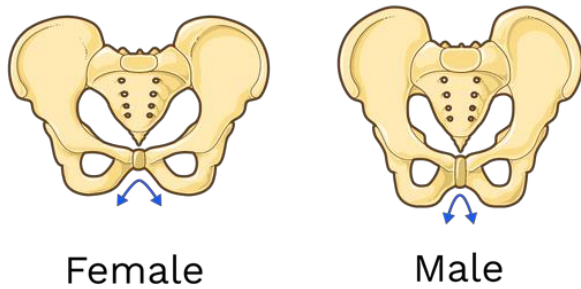


Bipedalism

Why a biped?

Consequences

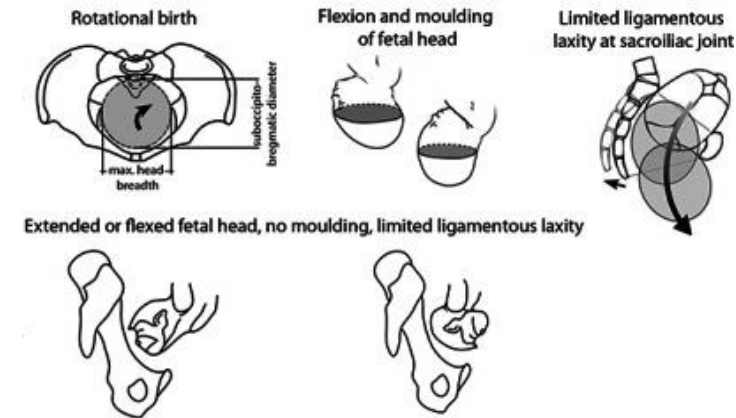
Bipedalism reorganizes the entire postcranial skeleton ...which entails huge tradeoffs



Sexual dimorphism in morphology

Life history

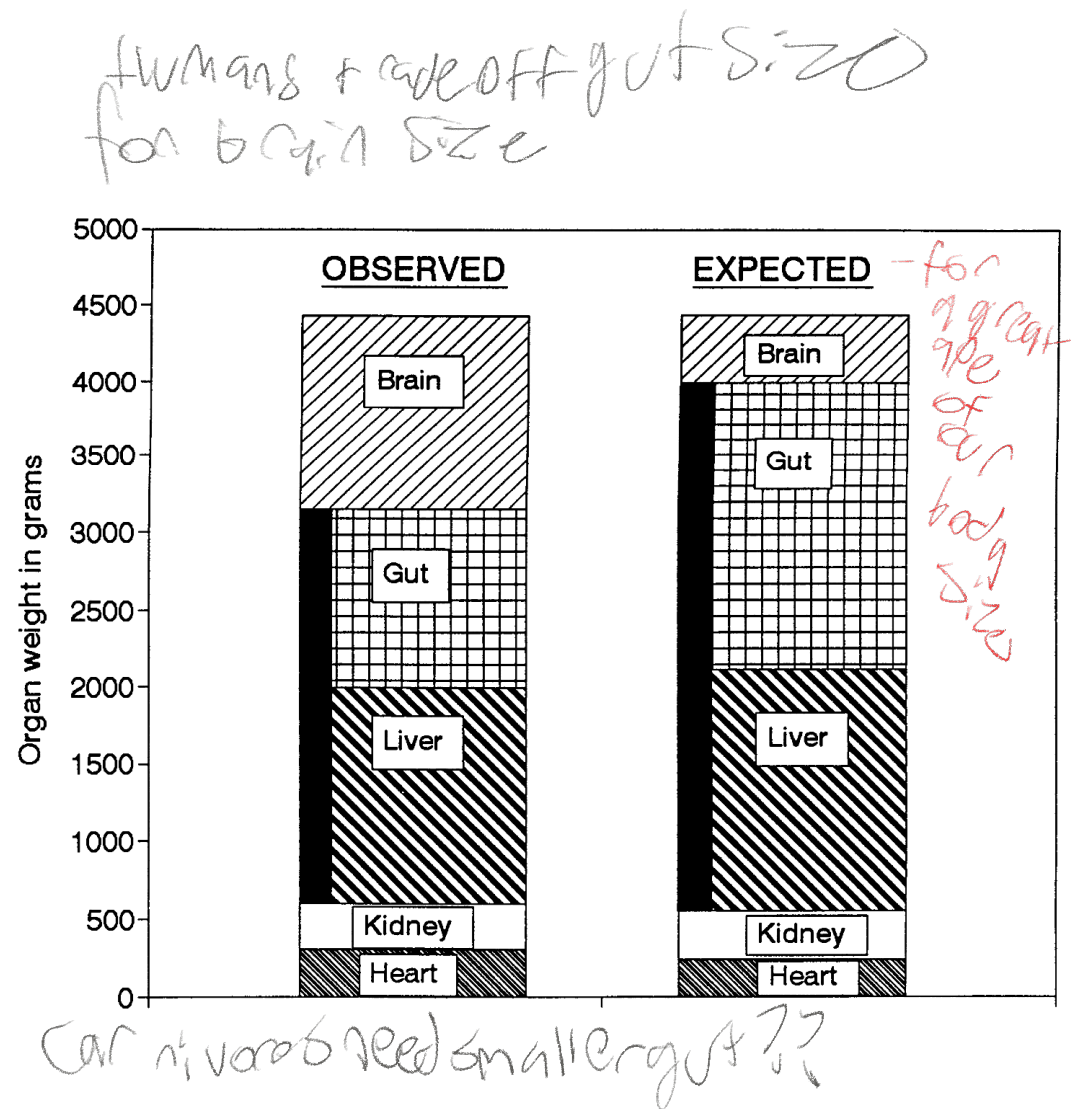
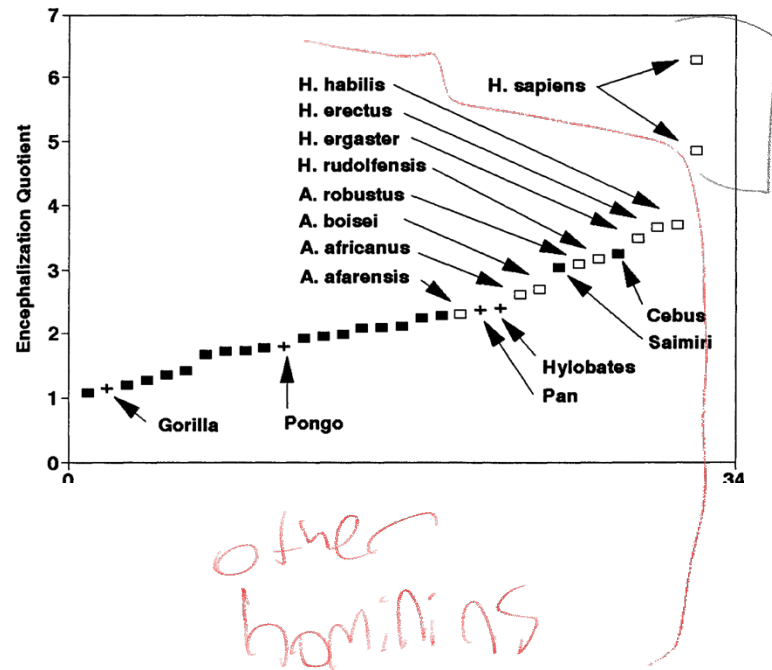
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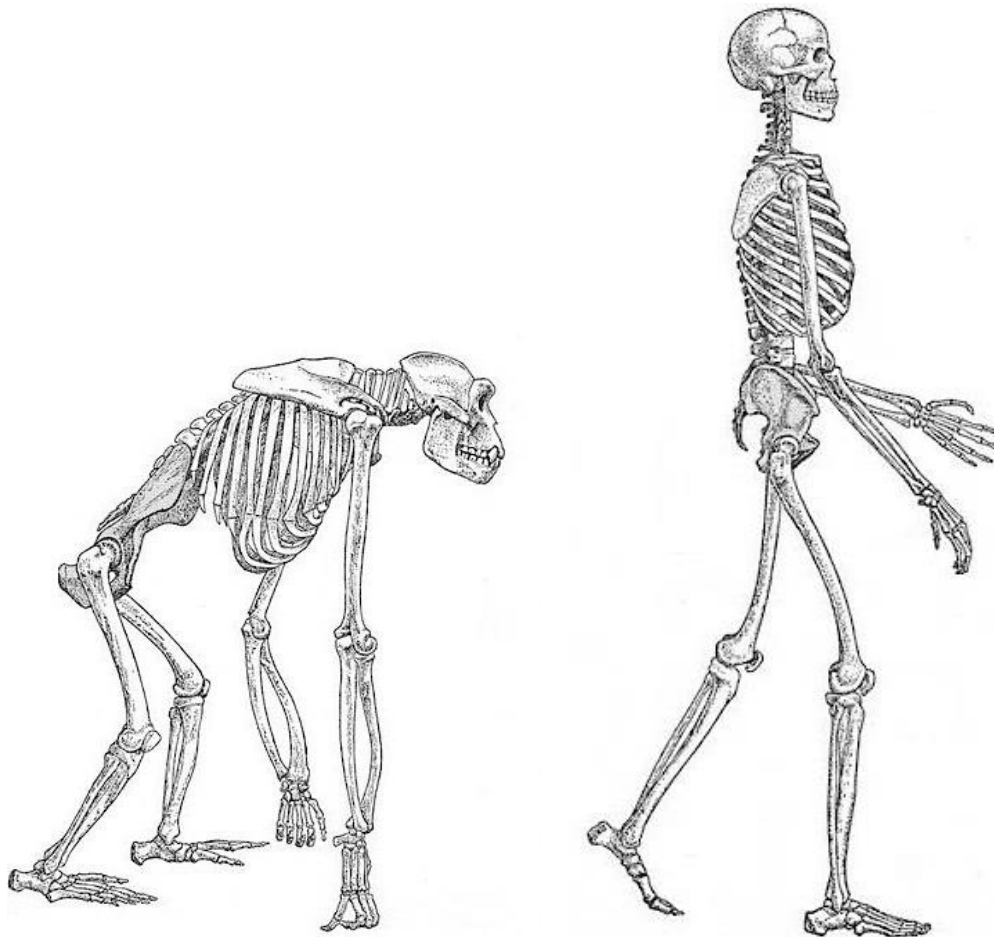
Birth physiology

...and associated social structures

Speaking of tradeoffs...



Bipedalism reorganizes the entire postcranial skeleton



EFFICIENT BIPEDAL LOCOMOTION

Foramen magnum inferoanteriorly placed

Shoulder Scapulae oriented so that humerus on side

Arms Short, relative to body and legs

Lumbar spine Flexible; lordosis (S-shaped spine)

Pelvis 'Bowl-shaped'. Ilium supero-inferiorly shortened and sagittally oriented. Mediolaterally broad.

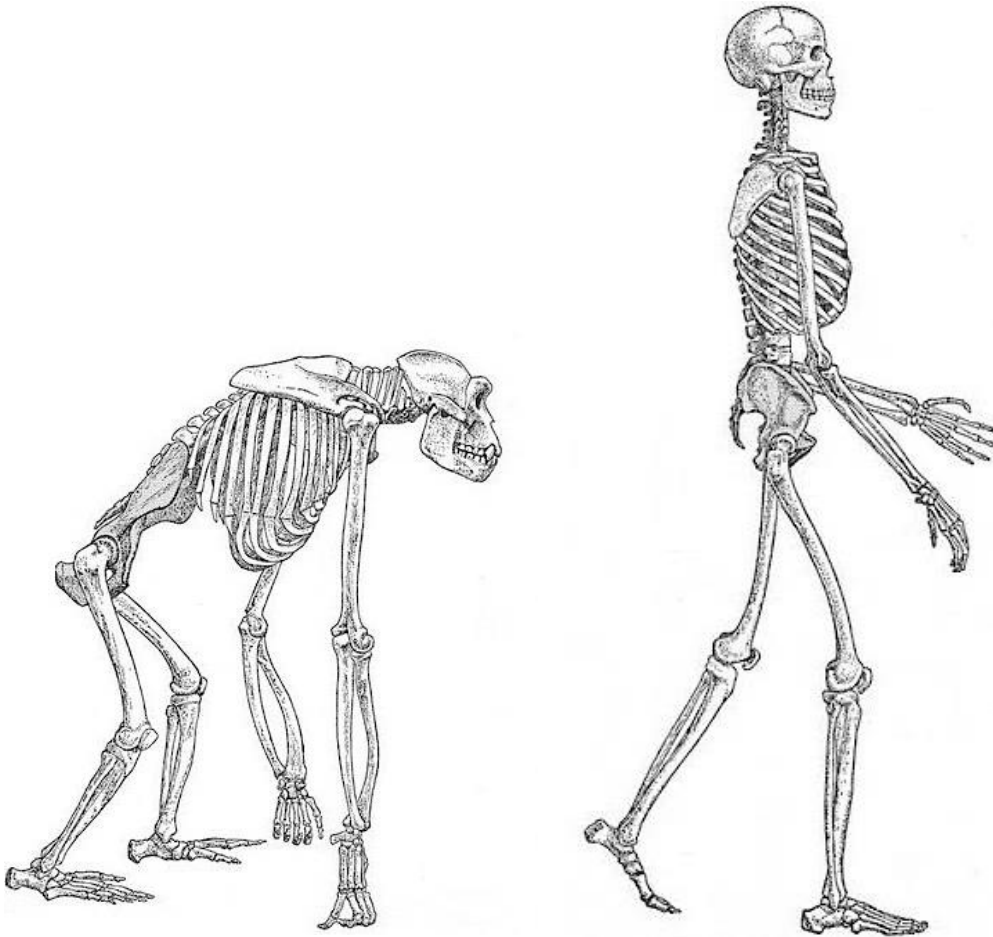
Legs Long relative to body and arms. Q and valgus angle

Leg joints Larger than in chimpanzees

Feet Stiff midfoot; large calcaneus; adducted hallux

Split from Neanderthals

Bipedalism reorganizes the entire postcranial skeleton



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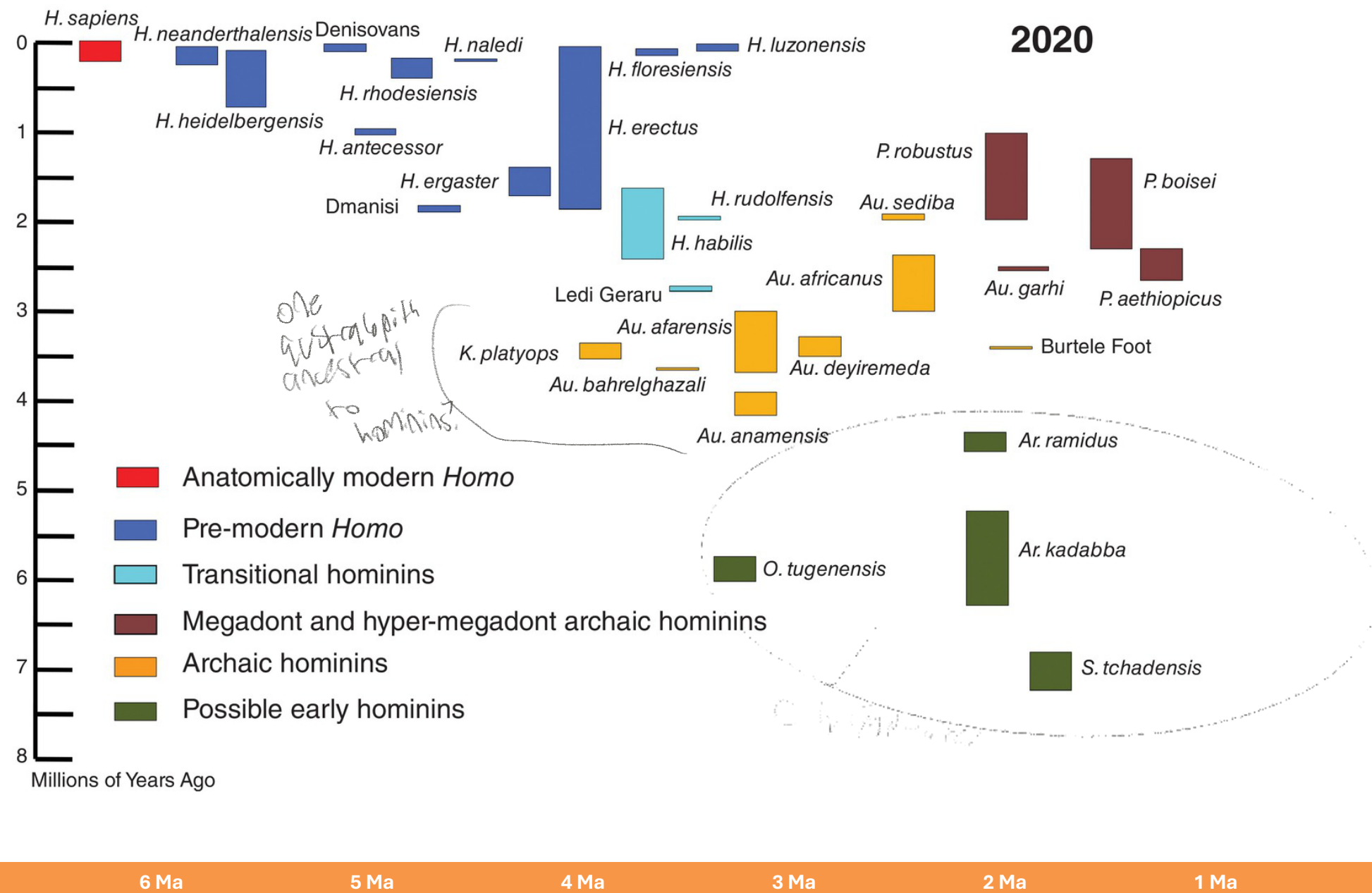
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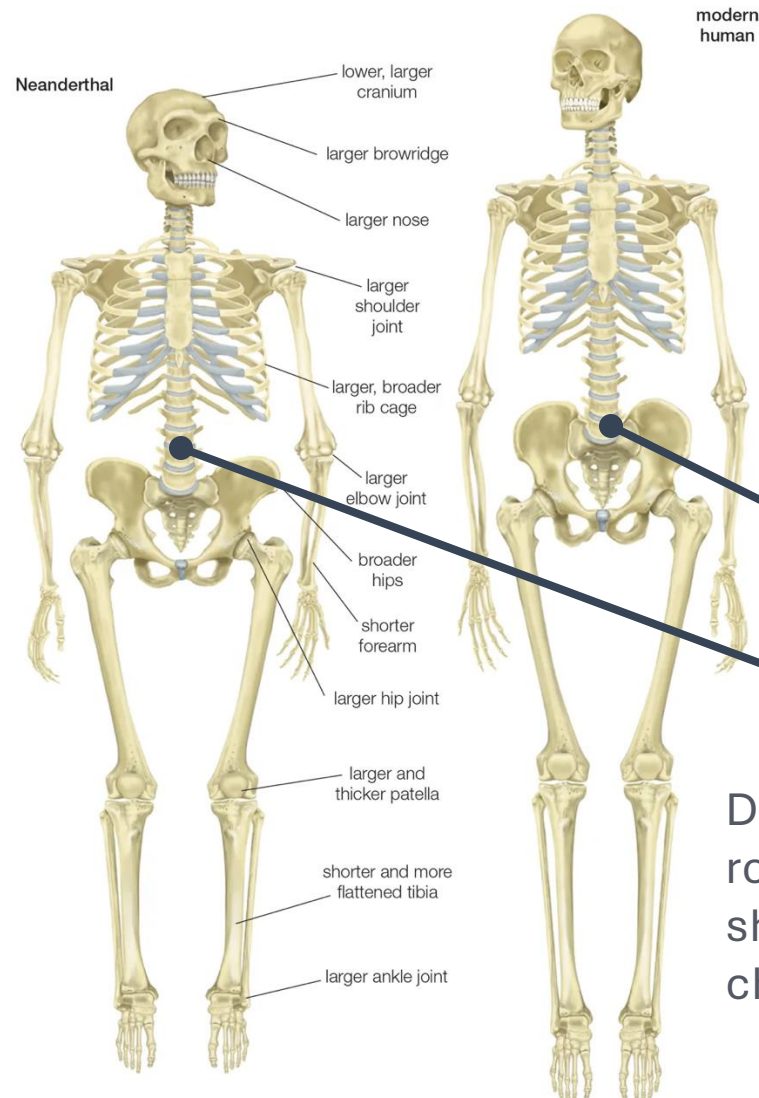


Bipedalism

Why a biped?

Consequences

The establishment of efficient bipedal locomotion



Differences in robusticity, but shared suite of characters

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Feet Stiff midfoot; large calcaneus; adducted hallux

7 Ma

6 Ma

5 Ma

4 Ma

3 Ma

2 Ma

1 Ma

The establishment of efficient bipedal locomotion



Sima de los Huesos; 0.4 Ma
Homo heidelbergensis

EFFICIENT BIPEDAL LOCOMOTION

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The establishment of efficient bipedal locomotion

Turkana boy; 1.5 Ma
Homo erectus



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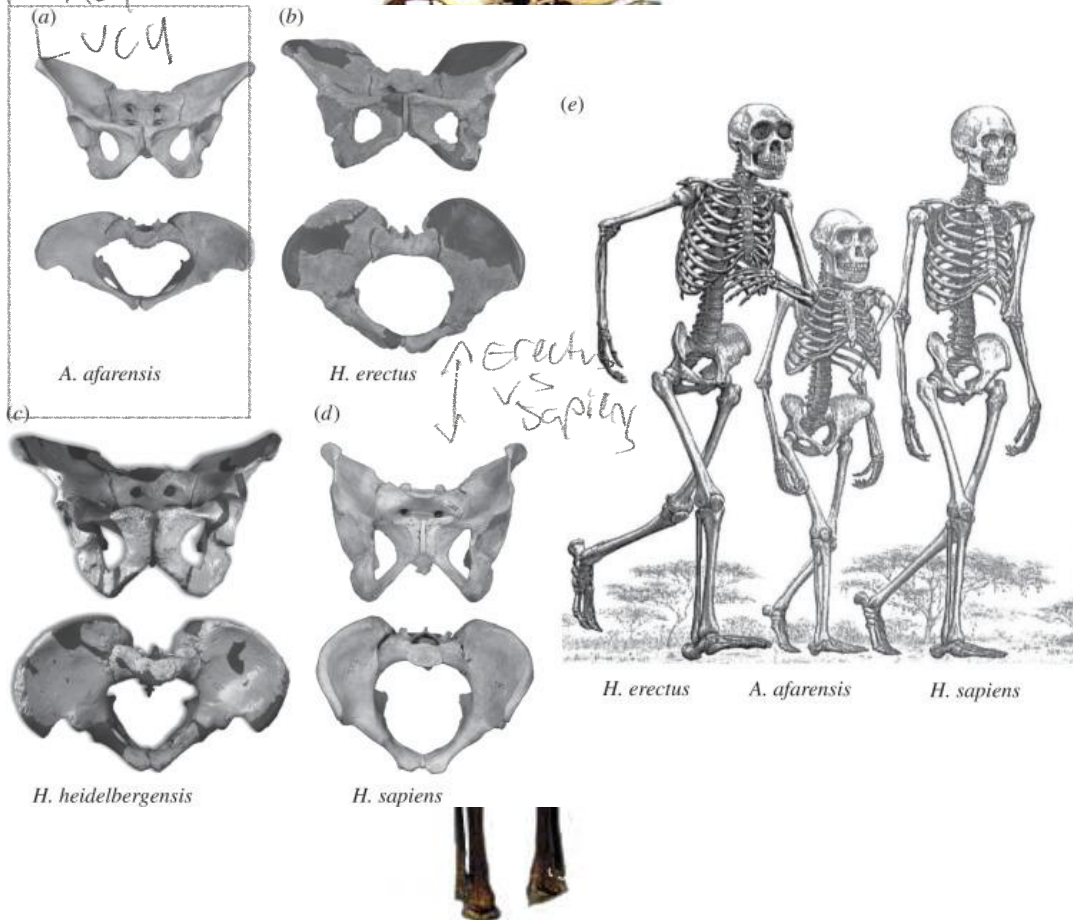
2 Ma

1 Ma

The establishment of efficient bipedal locomotion

Turkana boy; 1.5 Ma

Homo erectus



EFFICIENT BIPEDAL LOCOMOTION

What - like

Foramen magnum inferoanteriorly placed

Shoulder Scapulae: humerus oriented forwards

Arms Short, relative to body and legs

Lumbar spine Flexible; lordosis (S-shaped spine)

Pelvis Retains early form (**mediolaterally broad** and anteroposteriorly narrow, and flaring iliac blades).

Legs Long relative to body and arms. Q and valgus angle

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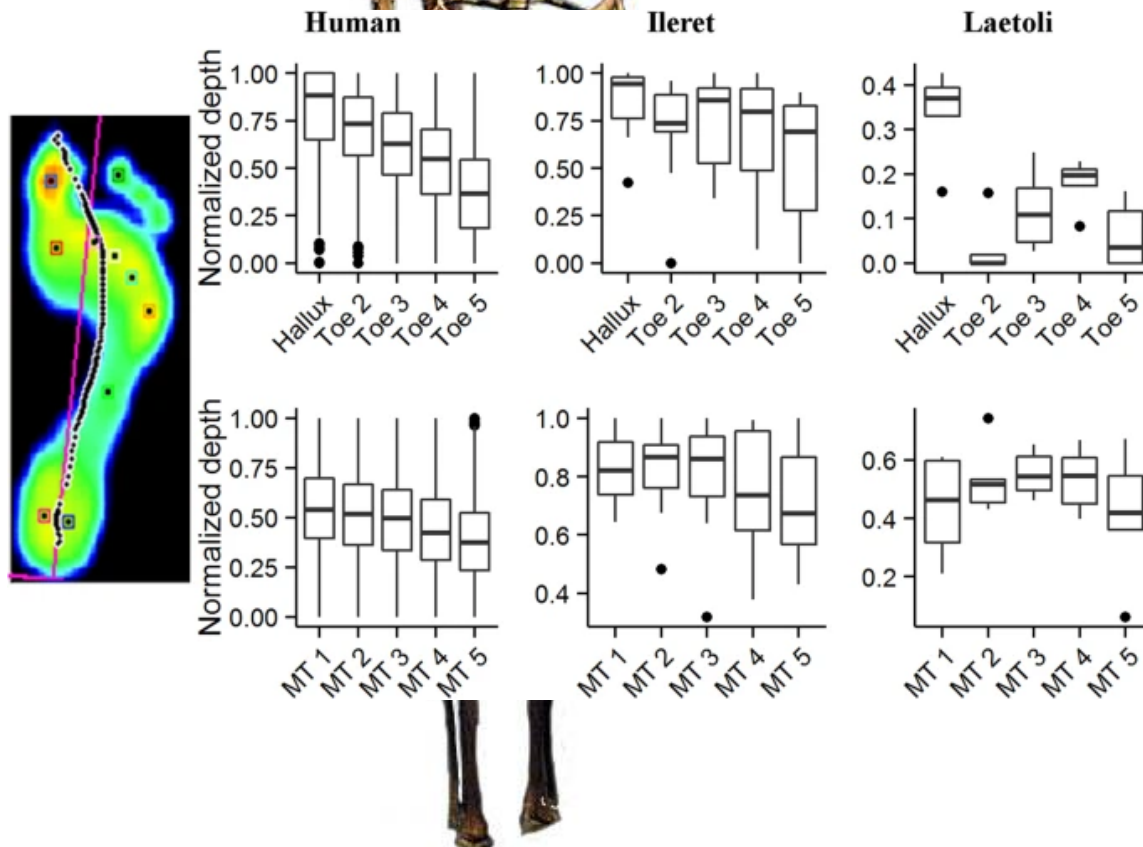
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The establishment of efficient bipedal locomotion

Turkana boy; 1.5 Ma
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1 Ma

The establishment of efficient bipedal locomotion

HUMAN-LIKE BIPEDAL LOCOMOTION

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Leg joints Larger than in chimpanzees

Feet Stiff midfoot; large calcaneus; adducted hallux

*Human like
bipedalism by
2 Ma*

7 Ma

6 Ma

5 Ma

4 Ma

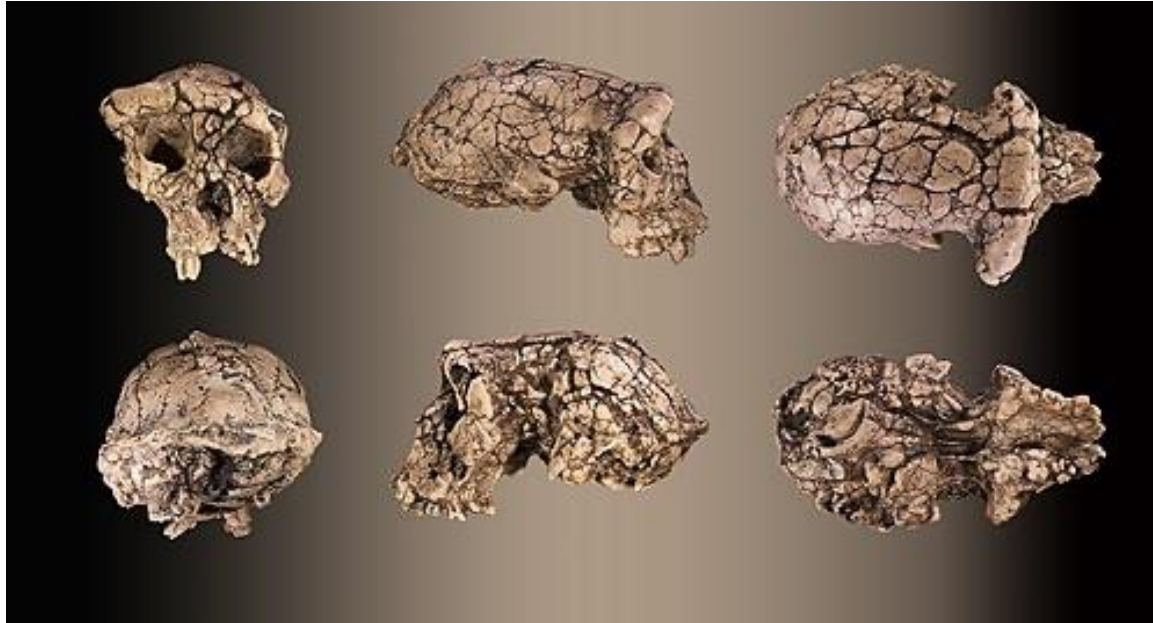
3 Ma

2 Ma

1 Ma

The establishment of efficient bipedal locomotion

Sahelanthropus - something transitional



Foramen magnum inferoanteriorly placed (!)

Arms Indicate knuckle-walking

Legs Inconclusive evidence

BIPEDAL LOCOMOTION

7 Ma

6 Ma

5 Ma

4 Ma

3 Ma

2 Ma

1 Ma

The establishment of efficient bipedal locomotion



Foramen magnum inferoanteriorly placed

Shoulder Ape-like, adapted for hanging

Arms Long (in some species)

Pelvis Mediolaterally broad and anteroposteriorly narrow, and flaring iliac blades).

Legs Long relative to body and arms. Q and valgus angle

Leg joints Larger than in chimpanzees

Feet Adapted for bipedalism (in some species)

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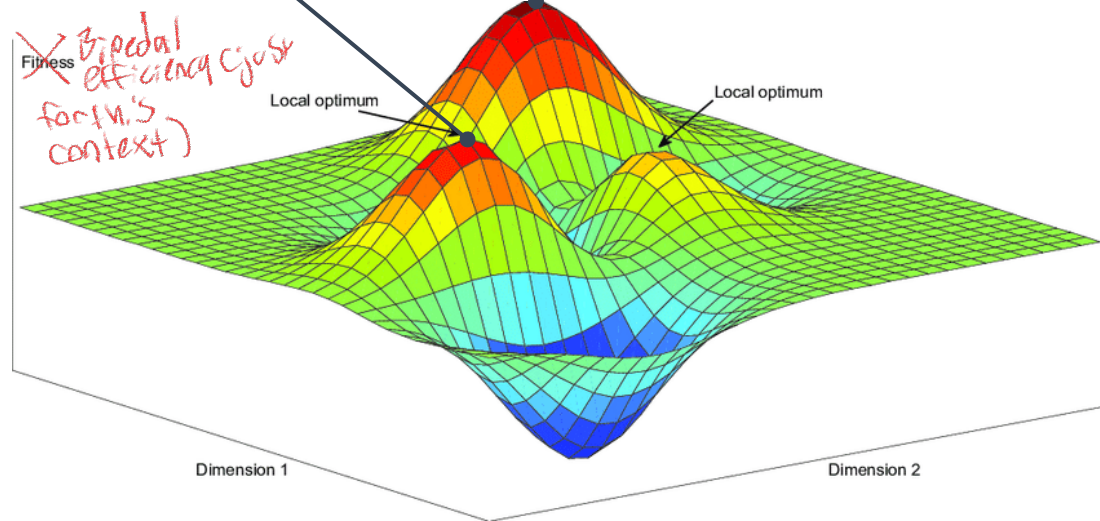
4 Ma

3 Ma

2 Ma

1 Ma

One way to think about this...



Not restricted
to bipedalism
can also
climb +
live in trees

DIVERSE FORMS OF
BIPEDAL LOCOMOTION

HUMAN-LIKE
BIPEDAL LOCOMOTION

7 Ma

6 Ma

5 Ma

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What's the difference?

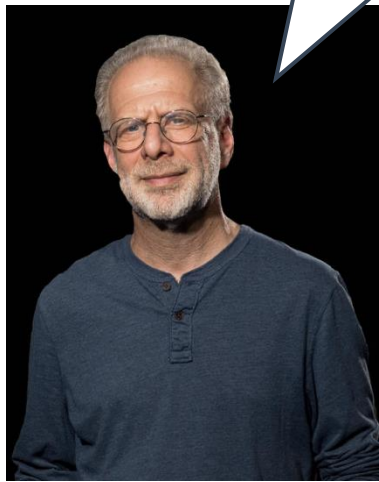
It's not walking efficiency
that sets humans apart,
it's running

Running!



David Carrier

What he said



Dan Lieberman

DIVERSE FORMS OF
BIPEDAL LOCOMOTION

HUMAN-LIKE
BIPEDAL LOCOMOTION

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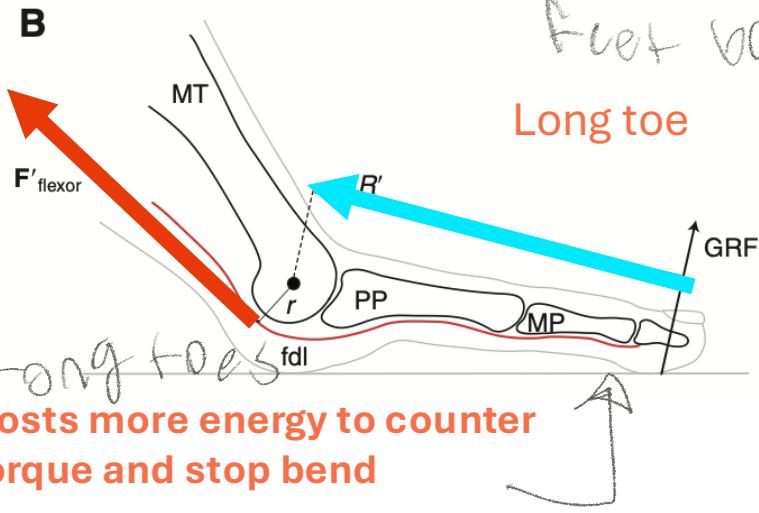
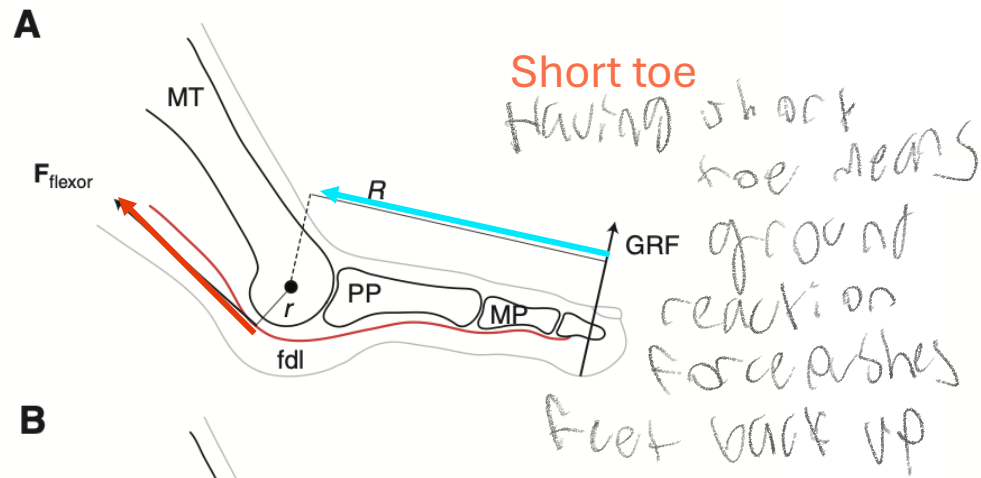
2 Ma

1 Ma

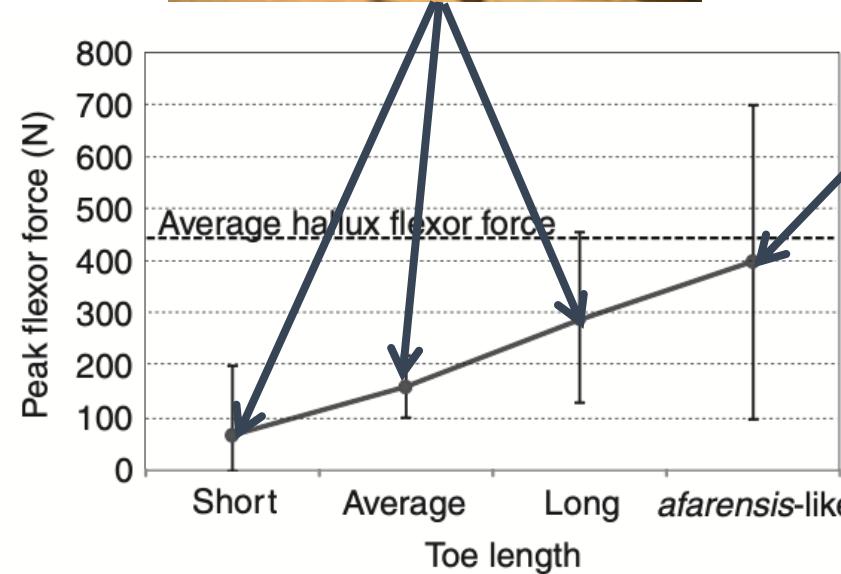
Long-distance running

Humans have short toes for efficient bipedal walking

running - short toes mean expending less energy



to exert force



Toe length (not) displayed here is a hypothetical exaggerated

In running, "...modern toes that are only 20% longer than average require peak forces and impulses that are nearly twice as large..."

Toe length doesn't make difference for walking but does for running

HUMAN-LIKE BIPEDAL LOCOMOTION

Rolian et al. (2009) J. Exp. Biology

DIVERSE FORMS OF BIPEDAL LOCOMOTION

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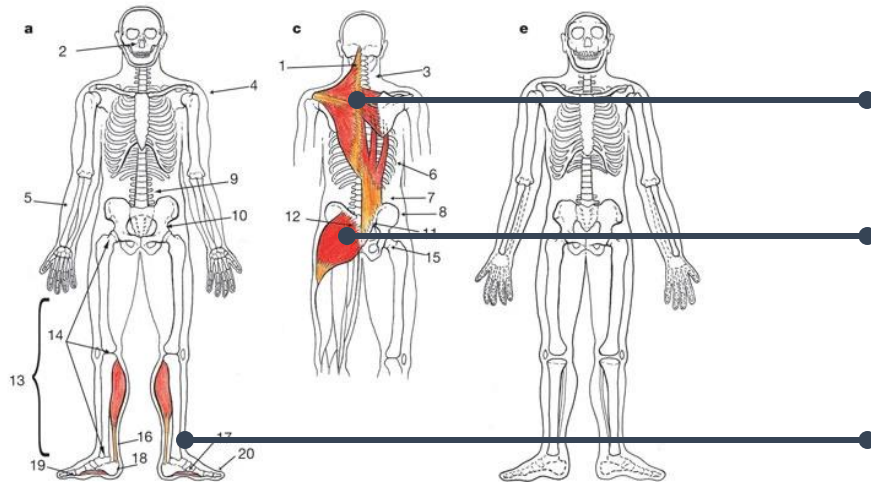
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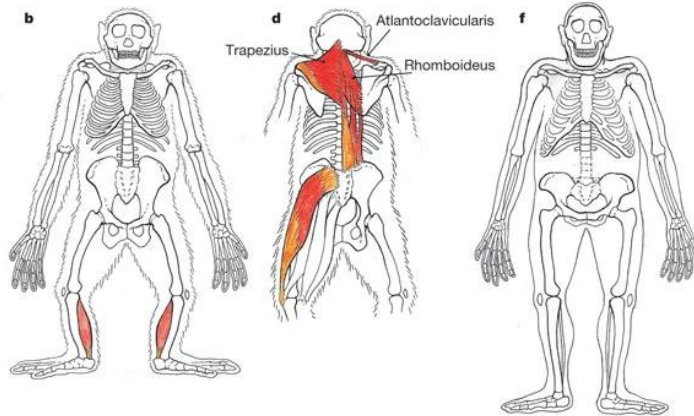
Long-distance running: adaptations for staying upright



Reduced muscles in shoulder: allows core to move in opposing direction to hips

Gluteus maximus: recruited during running, not walking (on flat surfaces)

Elastic storage in tendons (Achilles tendon)



+ Physiological adaptations
(e.g., sweating, thorax expansion)

DIVERSE FORMS OF
BIPEDAL LOCOMOTION

HUMAN-LIKE
BIPEDAL LOCOMOTION

Bramble & Lieberman. (2004) Nature

7 Ma

6 Ma

5 Ma

4 Ma

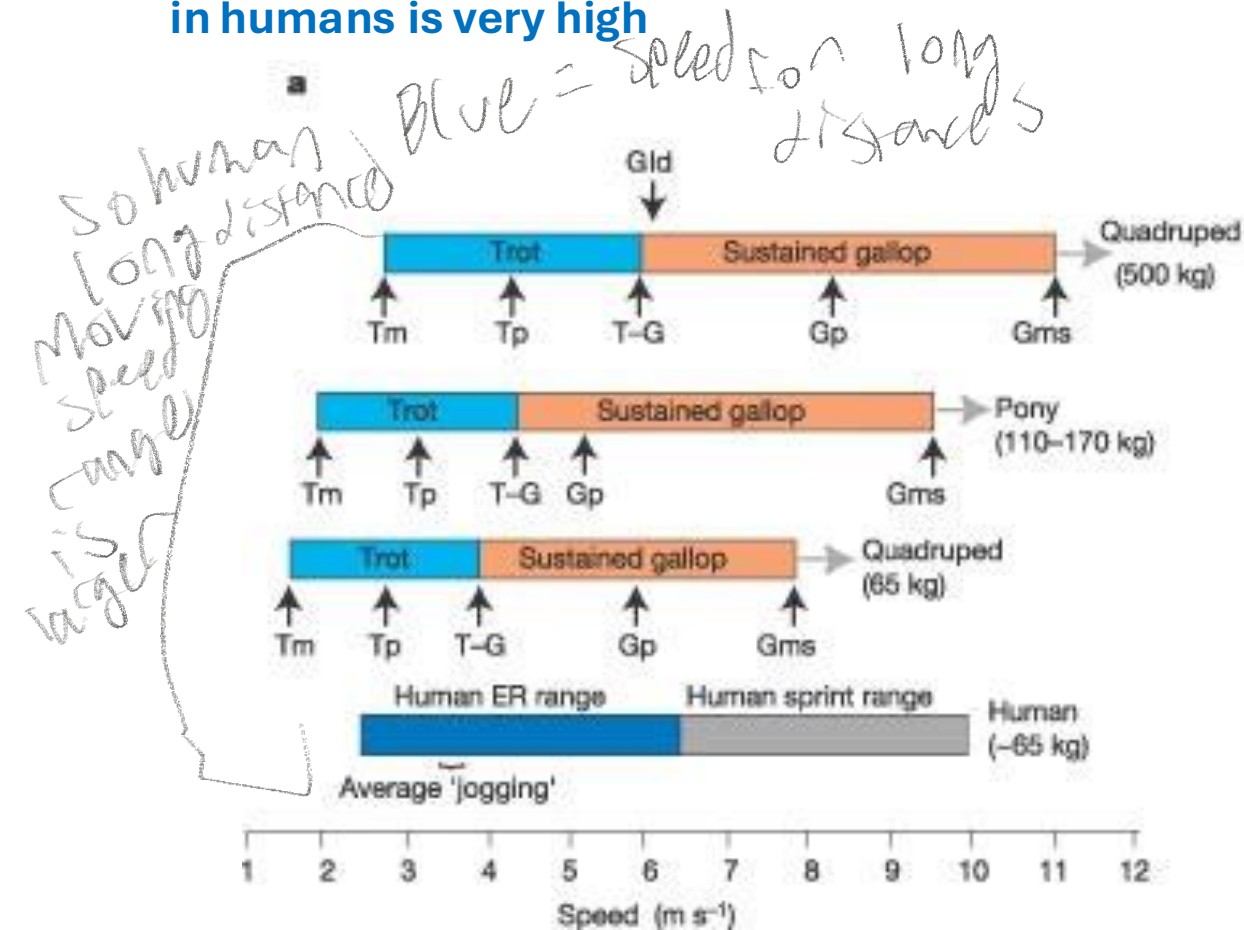
3 Ma

2 Ma

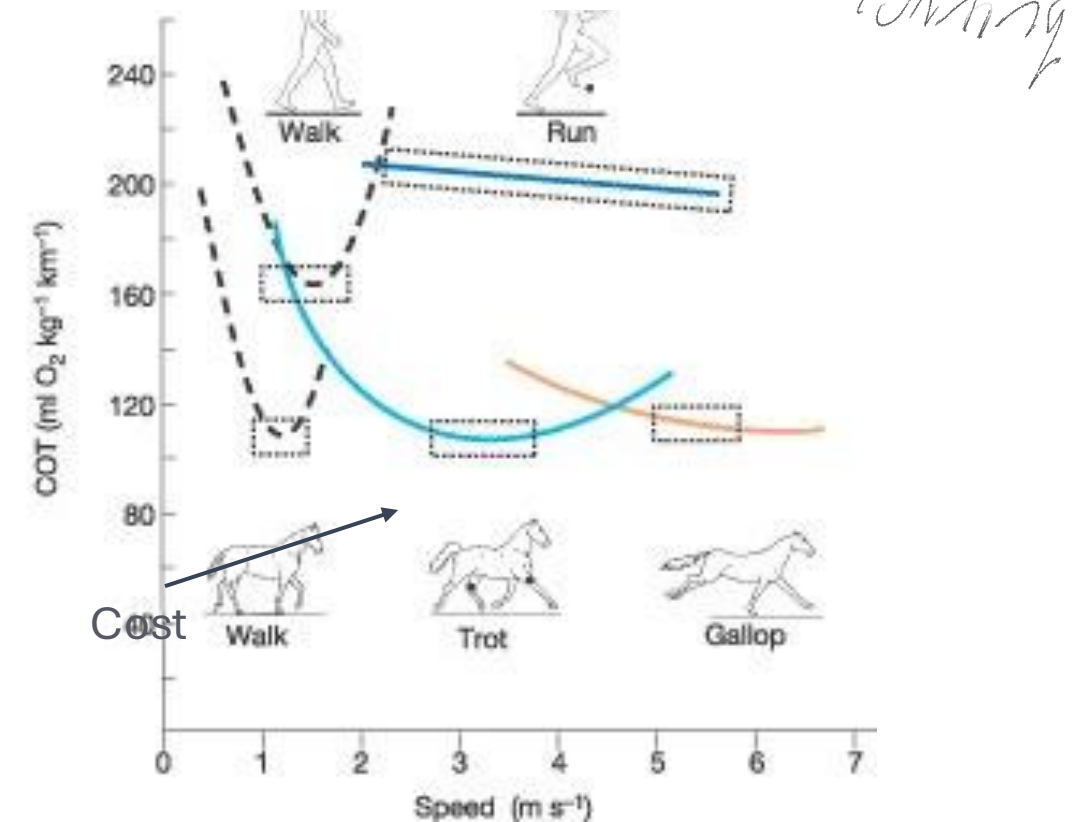
1 Ma

Humans are very good long-distance runners!

Speed at which long distances can be run in humans is very high



The cost of running faster does not increase, as it does in other long-distance runners:

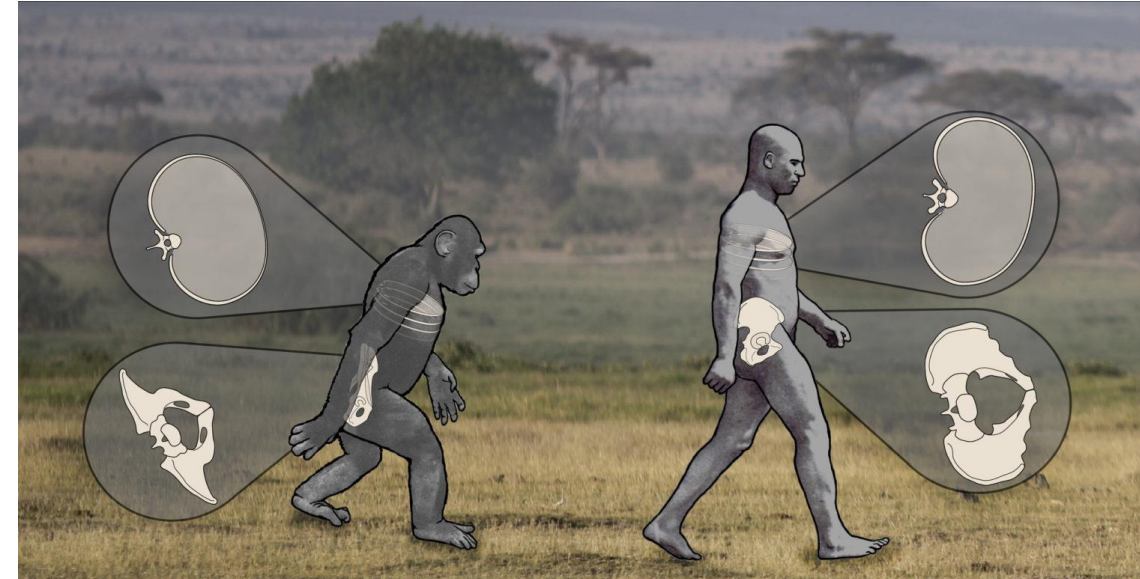


Long-distance running and throwing



Coevolution between throwing and running?

*Humeral adaptation
to running for
priming arms for big
range of motion for
arms - easier to
in cowfing? (heavens)*



Decoupled hips and thorax permits torso rotation:
needed for throwing

DIVERSE FORMS OF
BIPEDAL LOCOMOTION

HUMAN-LIKE
BIPEDAL LOCOMOTION

Roach et al. (2013) Nature

7 Ma

6 Ma

5 Ma

4 Ma

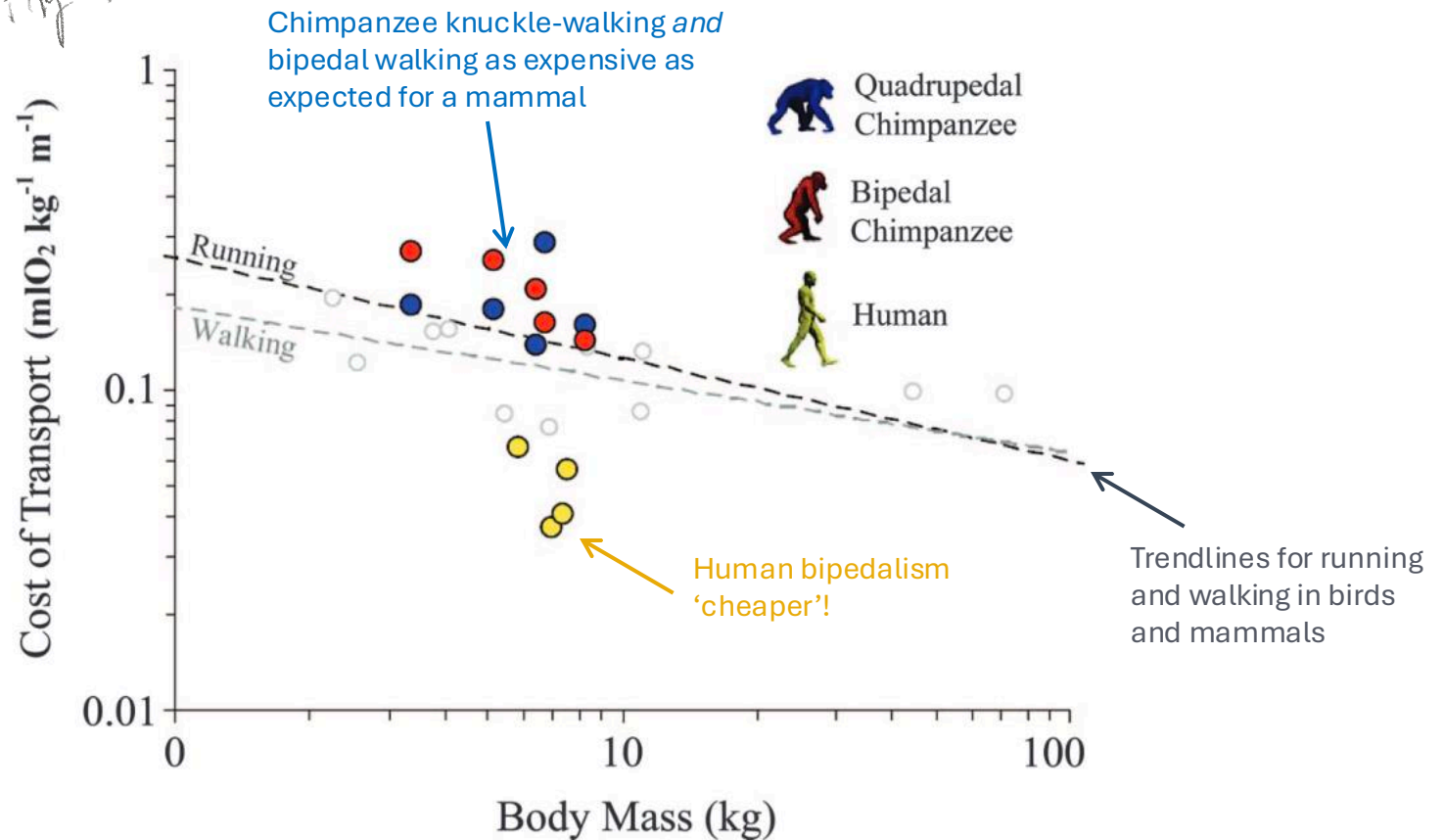
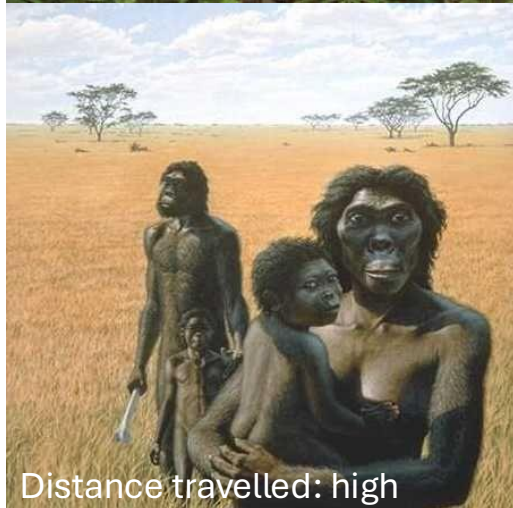
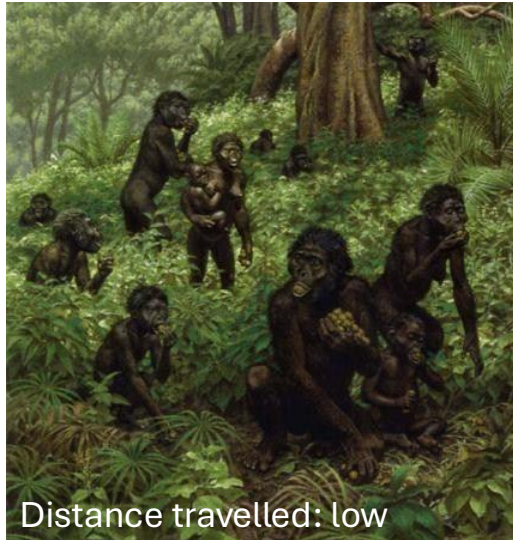
3 Ma

2 Ma

1 Ma

Bipedalism

Why is there bipedalism just for walking not running?



DIVERSE FORMS OF
BIPEAL LOCOMOTION

HUMAN-LIKE
BIPEAL LOCOMOTION

Sockol et al. (2007) PNAS

7 Ma

6 Ma

5 Ma

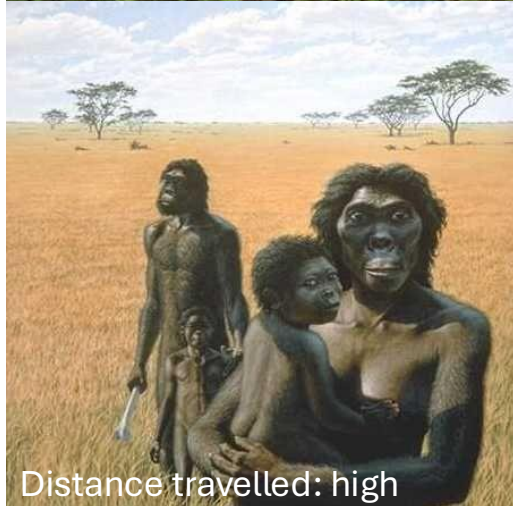
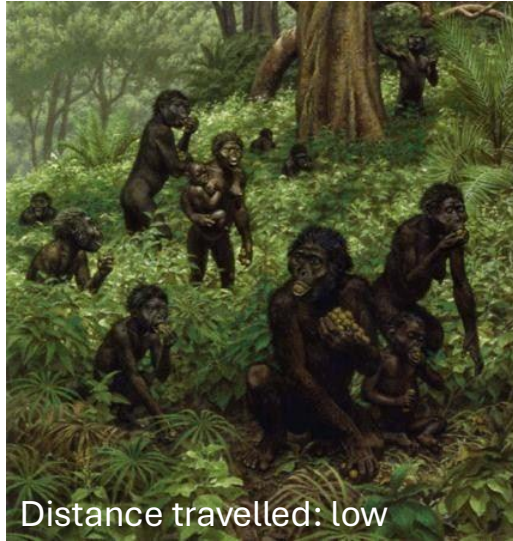
4 Ma

3 Ma

2 Ma

1 Ma

Bipedalism



DIVERSE FORMS OF
BIPEDAL LOCOMOTION



HUMAN-LIKE
BIPEDAL LOCOMOTION

7 Ma

6 Ma

5 Ma

4 Ma

3 Ma

2 Ma

1 Ma

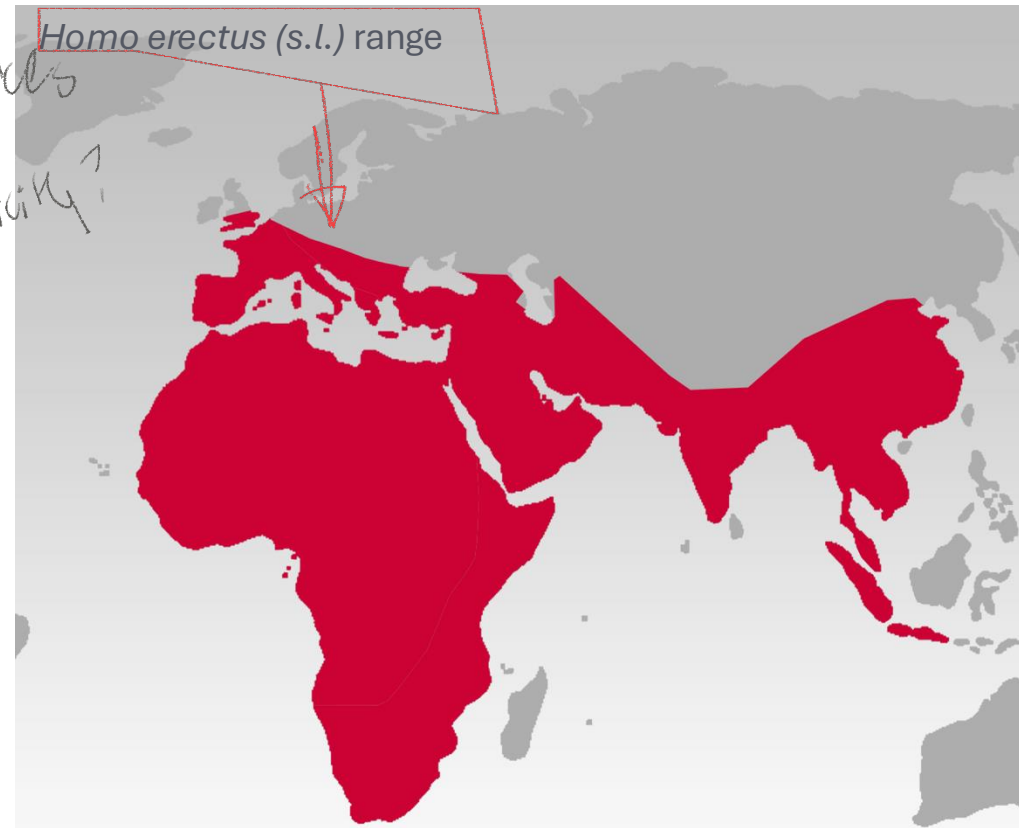
Did *Homo erectus* adapt to new environments through plasticity?

Large spread of *H. erectus* results in ability to move long distances

Modern bipedalism permits larger ranges & range expansions: exposure to new stressors



Leads to plasticity?



Secondary altriciality necessitates longer postnatal growth: permits 'more' developmental plasticity



The Biology of the Colonizing Ape

Jonathan C.K. Wells^{1*} and Jay T. Stock²

7 Ma

6 Ma

5 Ma

4 Ma

3 Ma

2 Ma

1 Ma

The modern human degree of plasticity *may* be established by 2 Ma



Various, including US immigrants and origin populations (relevant for controlling for genotype)

Looking for non-genetic differences with nutritional environment



Various wild vs captive and wild populations OWM (relevant for controlling for genotype...somewhat)

monkeys?

Is genotype actually controlled for?

pop's in different envs & all have different genotypes even if some

How do known local conditions covary with morphology?

- In **humans** and **OWMs**^{monkeys}, morphology is affected by environment
 - This pattern remains when controlling for sex & ancestry
- But... **non-human primates** show more robust responses to environmental change than humans do
 - Humans not so plastic after all?
 - Or an issue with samples?
 - Ancestry controlled for in humans
 - 'Pest' primates - not controlling for genotypes in monkey sample

Antón et al. (2016) *Phil Trans B*

Bipedalism

Why a biped?

Consequences

7 Ma

6 Ma

5 Ma

4 Ma

3 Ma

2 Ma

1 Ma

The modern human degree of plasticity *may* be established by 2 Ma



Various, including US immigrants and origin populations (relevant for controlling for genotype)



Various wild vs captive and wild populations OWM (relevant for controlling for genotype...somewhat)



Fossil hominins

- *Homo erectus*
- Neanderthals
- *Homo naledi*
- *Homo heidelbergensis*

Homo erectus:



Some matching attempted

- Dmanisi: more predators
- Zhoukoudian: fewer predators

Neanderthals:



Results:

- *Homo erectus* shows more variation between its 'populations' than Neanderthals
- *Very brief* mention of lower variation than modern humans

lots of confounders here

The modern human degree of plasticity *may* be established by 2 Ma



Various, including US immigrants and origin populations (relevant for controlling for genotype)



Various wild vs captive and wild populations OWM (relevant for controlling for genotype...somewhat)



Fossil hominins

- *Homo erectus*
- Neanderthals
- *Homo naledi*
- *Homo heidelbergensis*

My view: so many confounds

- *Are humans more or less plastic than other primates?*
 - Poor sampling strategy: in humans and nonhumans, this sample is not controlled for genetics
- Does *Homo erectus* respond plastically to its environment, thus buffering its genome from selection?
 - No natural experiment available to control for genetic adaptation
 - Environmental context poorly known
 - Some populations sampled span a long time; others don't

The samples 'just aren't there'

Antón et al. (2016) *Phil Trans B*

What about all the other traits?!

INVARIANT TRAITS THAT DEFINE HUMAN BIOLOGY

DNA	Metabolism	Sexual reproduction	Mitochondria	Central nervous system	Pentadactyly	Vertebrate jaw	Nails
RNA	Respiration		Muscle and nerve cells	Eyes		Mammalian ear	Stereoscopic vision
Cells	Glycolysis		Endoskeleton			Endothermy	Altriciality
ATP	Krebs cycle					Live birth	2.1.2.3
	Oxidative phosphorylation					Heterodonty	

Dead file
Saw has lots of
parts a mammal
Saw has 1
other parts of
mammal

Bipedalism

Why a biped?

Consequences

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ATP	Krebs cycle					2.1.2.3
	Oxidative phosphorylation					

REPTILE

Jaw joint: Quadrate (cranium) and Articular (mandible)



Mandible: multiple bones

MAMMAL

Jaw joint: Squamosal (cranium) and Dentary (mandible)



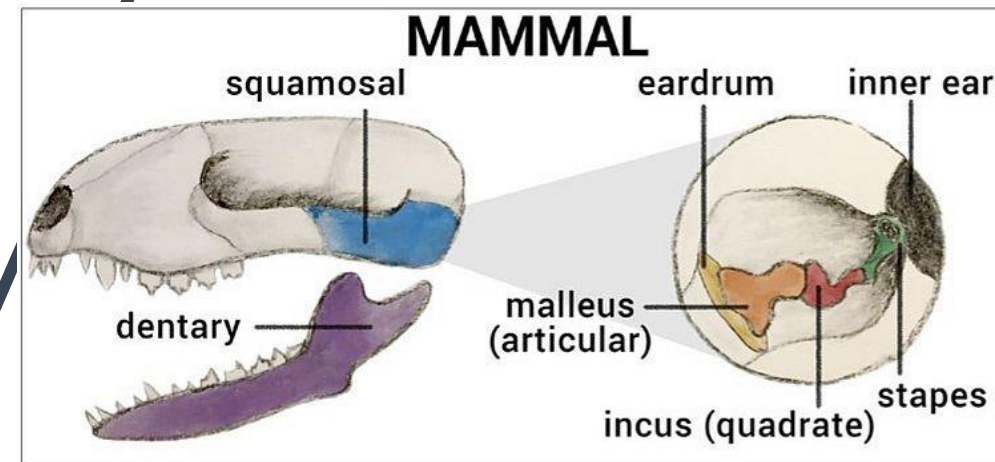
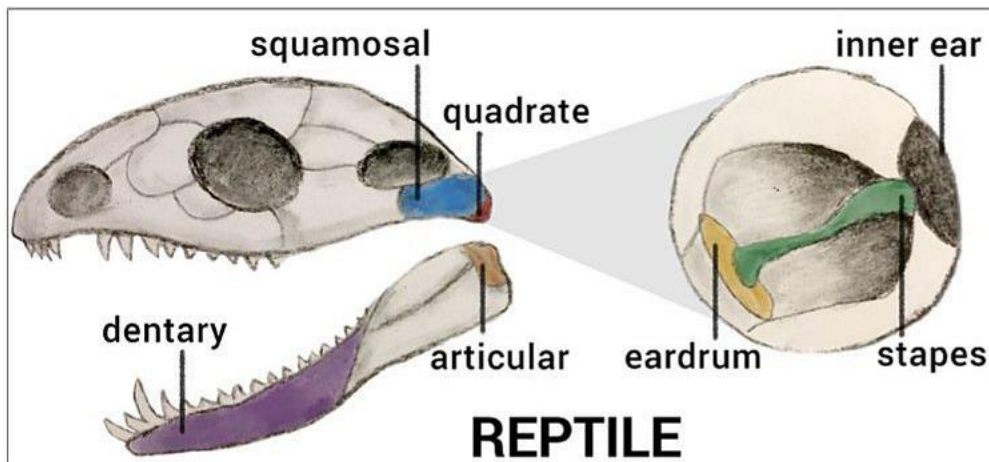
Mandible: single bone

Pentadactyly

What about all the other traits?!

INVARIANT TRAITS THAT DEFINE HUMAN BIOLOGY

DNA	Metabolism	Sexual reproduction	Mitochondria	Central	Pentadactyly	Vertebrate jaw	Nails
RNA	Respiration		Muscle and	nervous		Mammalian ear	Stereoscopic
Cells	Glycolysis		nerve cells	system		Endothermy	vision
ATP	Krebs cycle		Endoskeleton	Eyes		Live birth	Altriciality
	Oxidative phosphorylation					Heterodonty	2.1.2.3



Summary

- Modern human body proportions (themselves, of course, only a reorganization of the general vertebrate endoskeleton...) seem to be adaptations to **bipedal long-distance running**, and are established by 2 Ma
- These adaptations necessitate a total **reorganization** of the skeleton & entail **tradeoffs**, which to some extent are offset by an extended period of **postnatal growth**
- This period of growth, in combination with expanded ranges, *may* have resulted in the establishment of modern human levels of **plasticity** at 2 Ma
- You're listening with your jaws